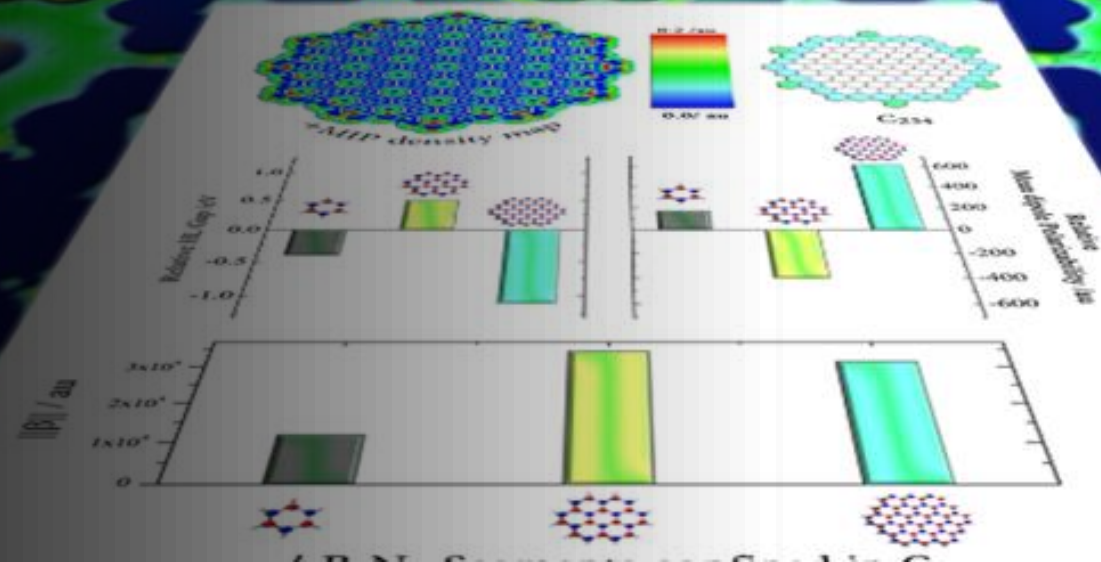
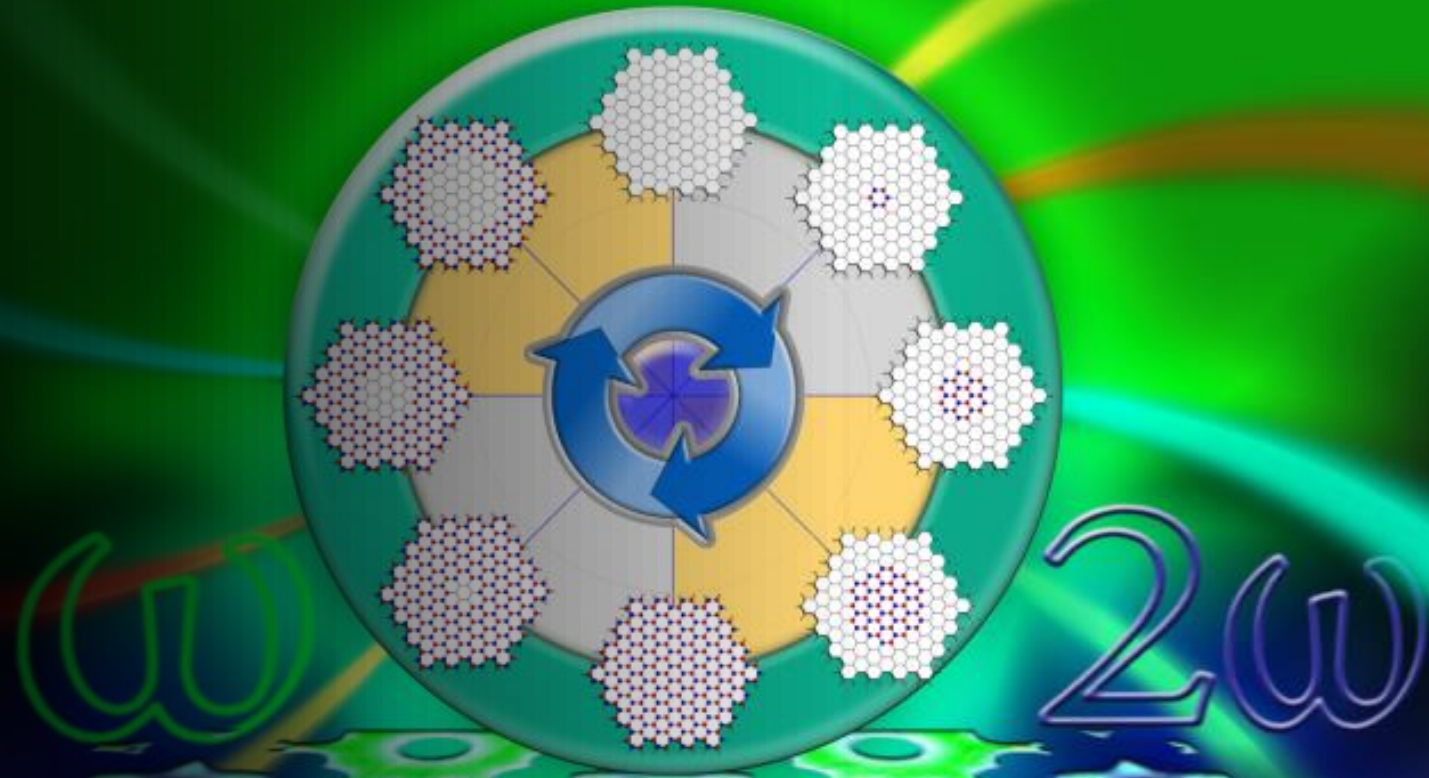




Beyond Linear Responses Exploring NLO Phenomena with CRYSTAL

Panaghiotis Karamanis

Centre National de la Recherche
Scientifique



Nonlinear optics is
present in our everyday
technologies

Non-Linear
Optics in our
pockets

Green Light
(visible)
532 nm`





Potassium Titanyl Phosphate

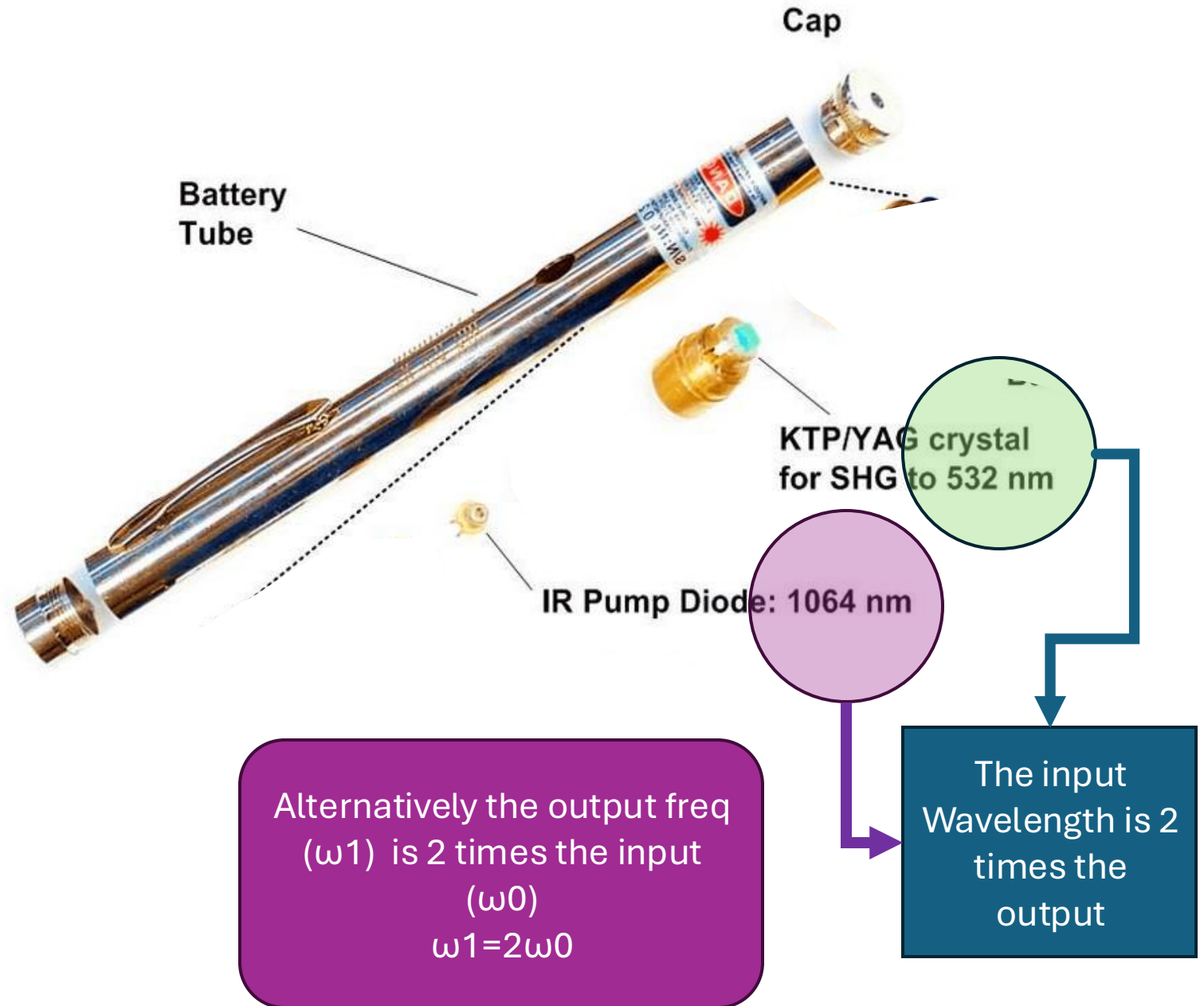
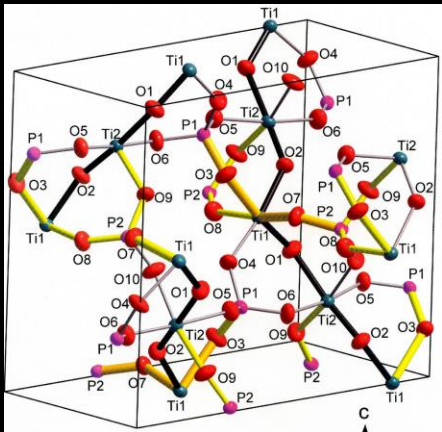
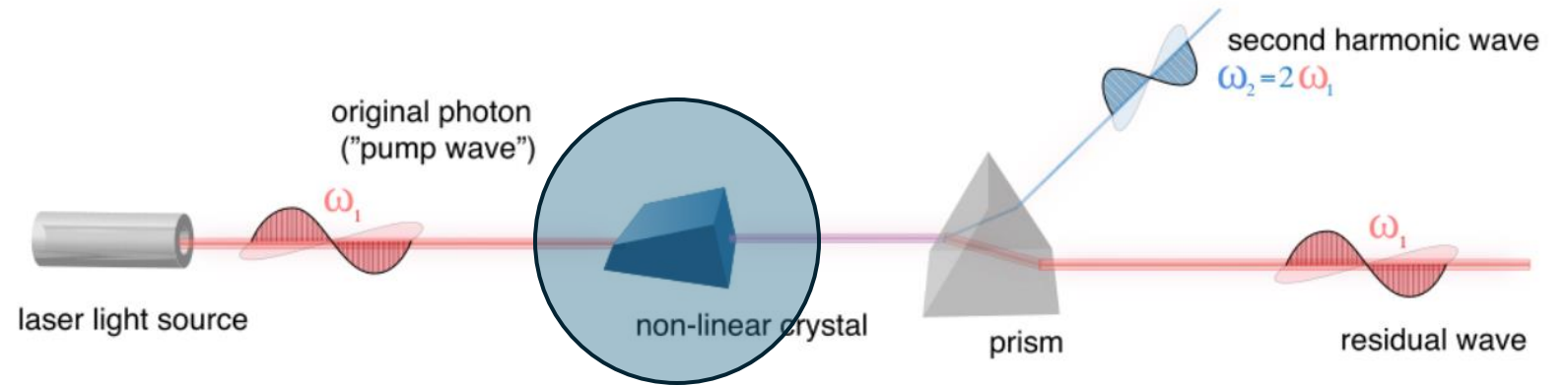


Image credit: Karlsruher Institut für Technologie

SECOND Harmonic Generation

Light can change its color if it
interacts with the **right material**

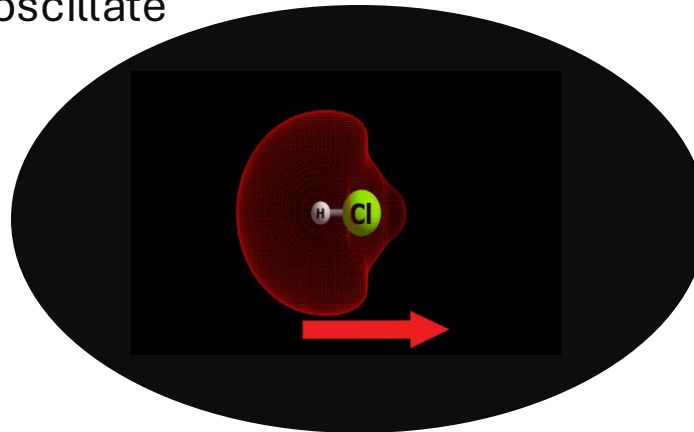


Nonlinear optics is not
about Light alone, but also
about **Materials**

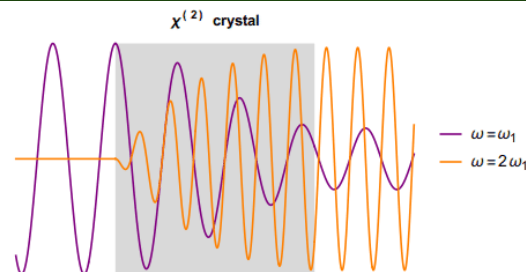
Classical View (oscillator model)

Atoms or molecules are modeled like **springs** (electrons bound to nuclei)

Light's electric field makes electrons to oscillate



Nonlinearities come from anharmonic oscillations of electrons.

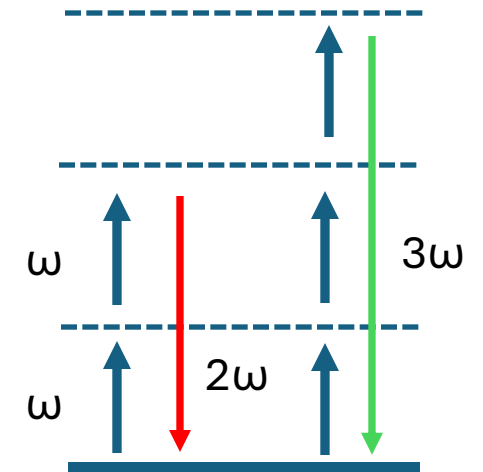


QM View

(Sum-Over-States / Virtual transitions)

Electrons occupy **discrete energy levels**

Light excites electrons to **virtual states**



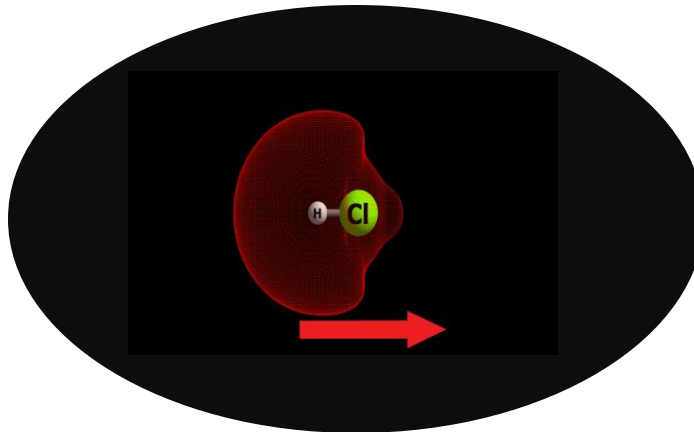
Nonlinearities come from multi-photon interactions via virtual states

Classical View (oscillator model)

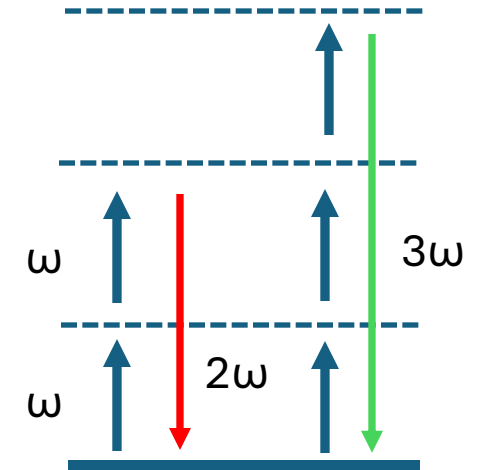
QM View (Sum-Over-States / Virtual transitions)

There are TWO
ways of
understanding
the origin of
NLO Effects

Despite the drastically different description of NLO



Anharmonic oscillations of electrons.



Multi-photon interactions via virtual states

Classical View (oscillator model)

QM View (Sum-Over-States / Virtual transitions)

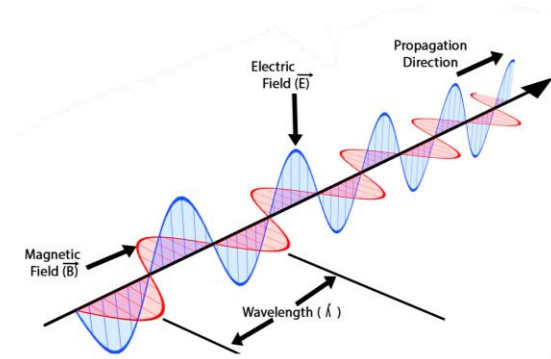
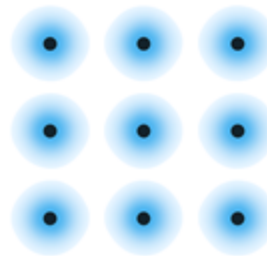
There are TWO
ways of
understanding
the origin of
NLO Effects

Both Approaches converge to the same physical effect

**Polarization of the electron density
via a Time-Dependent Electrical Perturbation**

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

Electron density



Classical View (oscillator model)

QM View (Sum-Over-States / Virtual transitions)

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

Which in turn gives rise to....

To the :

✓ **Linear**

✓ **Nonlinear**

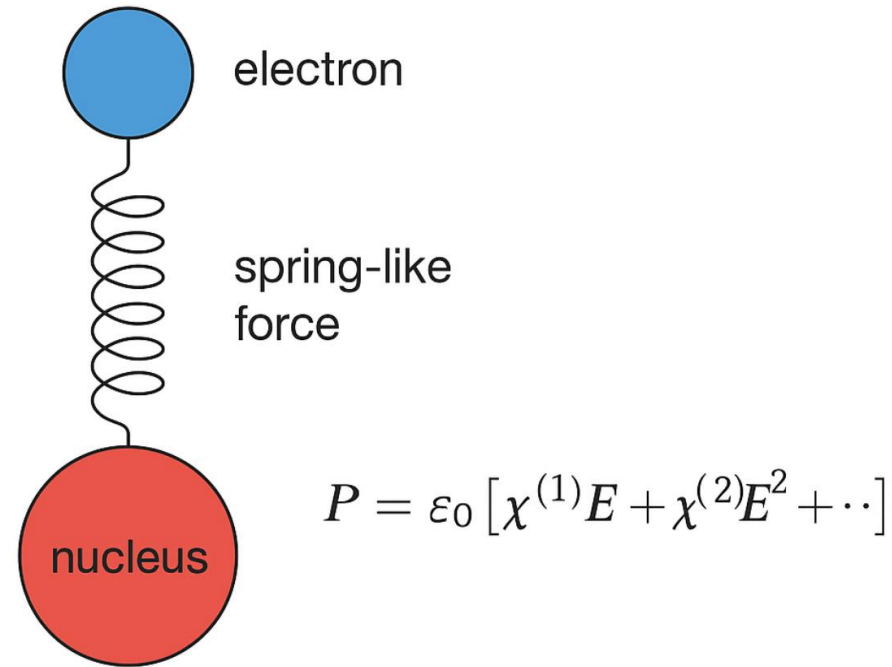
Susceptibilities

They come from
intrinsic electronic
features of the
material

Polarization Expansion Key points

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

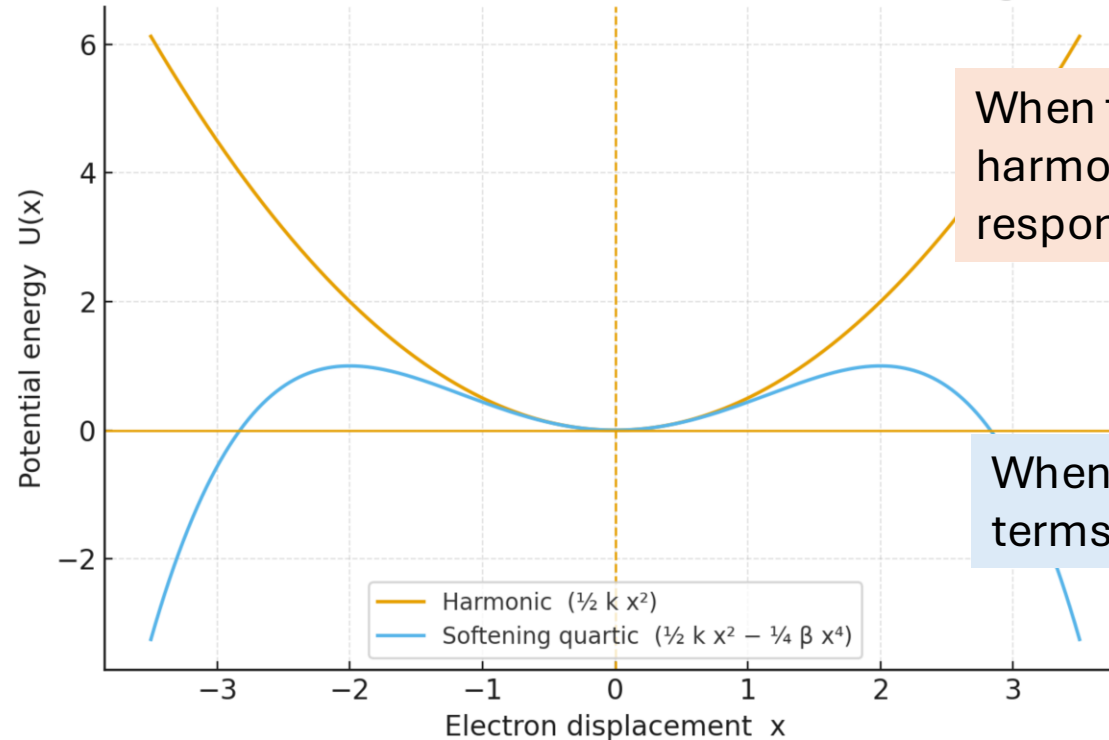
It has its roots in the **Lorentz oscillator model**



Polarization Expansion Key points

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

Lorentz Oscillator Potentials: Harmonic vs Softening Quartic



When the restoring force is harmonic, we get the linear response $\chi(1)$.

When it is anharmonic, higher-order terms naturally appear.

Polarization Expansion Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

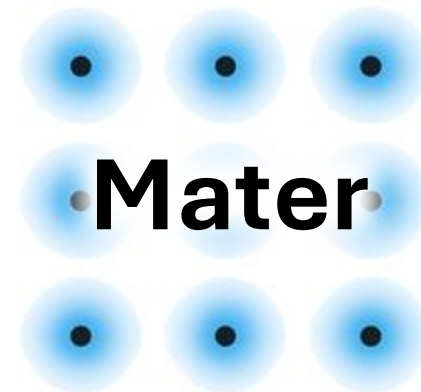
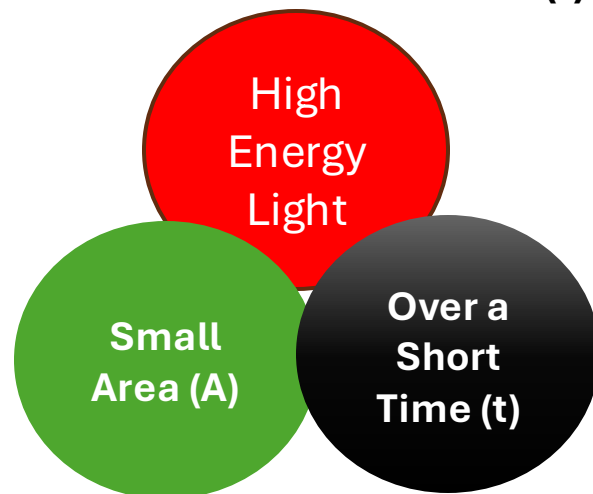
In light-matter NLO interaction, what truly matters is the light's intensity

$$I = \frac{P}{A}$$

I = **Intensity** of the light (W/m^2)
 P = **Power** of the light beam (W)
 A = **Area** over which the power is distributed (m^2)

$$P = \frac{\text{Light Energy } E(t)}{\text{time}}$$

$P(t)$ becomes Anharmonic only when...



Which means....



NLO is all about Lasers

Polarization Expansion Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(1)}$
Linear susceptibility

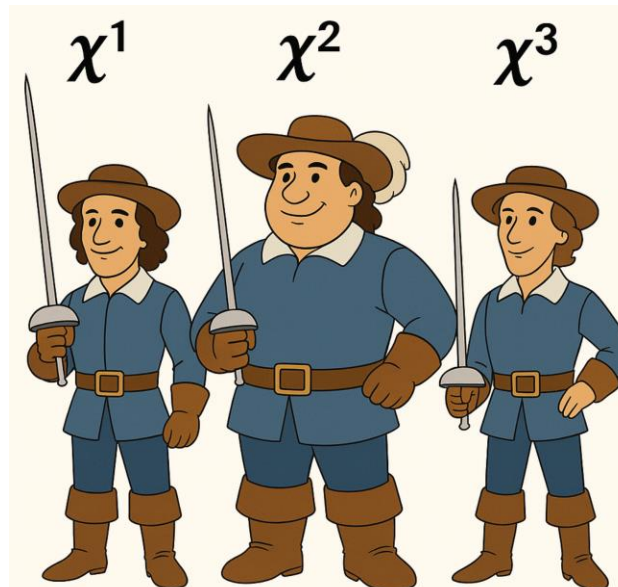
Athos: the **boring one**
✓ always present
✓ steady like linear optics.

$\chi^{(2)}$
Second-order
susceptibility

Porthos : the **flashy one**
✓ eye-catching effects like
SHG

$\chi^{(3)}$
Third-order
susceptibility

Aramis : the elegant and
versatile one
✓ appearing everywhere



Polarization Expansion Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(1)}$
Linear susceptibility

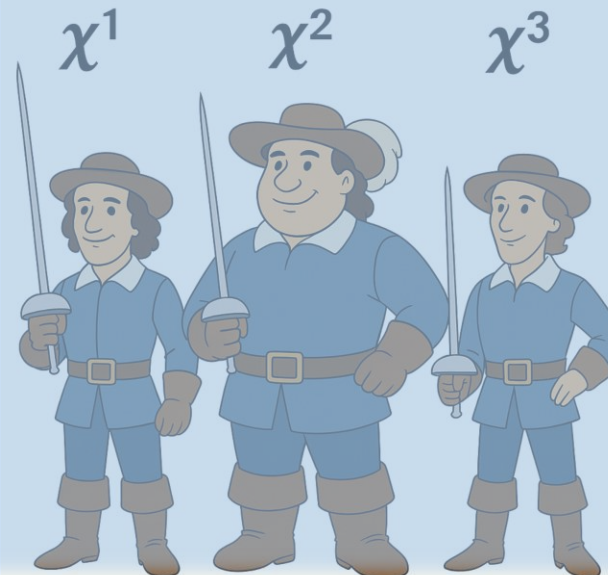
- Athos: the **boring one**
- ✓ always present
 - ✓ steady like linear optics.

$\chi^{(2)}$
Second-order
susceptibility

- Porthos : the **flashy one**
- ✓ eye-catching effects like SHG

$\chi^{(3)}$
Third-order
susceptibility

- Aramis : the elegant and **versatile one**
- ✓ appearing everywhere



who makes the
whole story unfold



Polarization Expansion

Key points

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(1)}$
Linear susceptibility

Measures how easily a material polarizes under a **weak** electric field

$$\mathbf{P}(t) = \epsilon_0 \chi^{(1)} \mathbf{E}(t) \quad \text{In a linear fashion}$$

Connected to:

$$\epsilon_r = 1 + \chi^{(1)} \quad n = \sqrt{1 + \chi^{(1)}}$$

Dielectric constant

Refractive index

$$n^2 = 1 + \chi^{(1)}$$

tells how strongly charges in the material respond to an electric field.

- Strong $\chi^{(1)}$ (high polarization) $\rightarrow$ large $n \rightarrow$ light goes slower
- Weak $\chi^{(1)}$ (low polarization) $\rightarrow$ small $n \rightarrow$ light goes faster

tells how fast light travels in the material.

Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(1)}$
Linear susceptibility

$\chi^{(1)}$ is generally complex

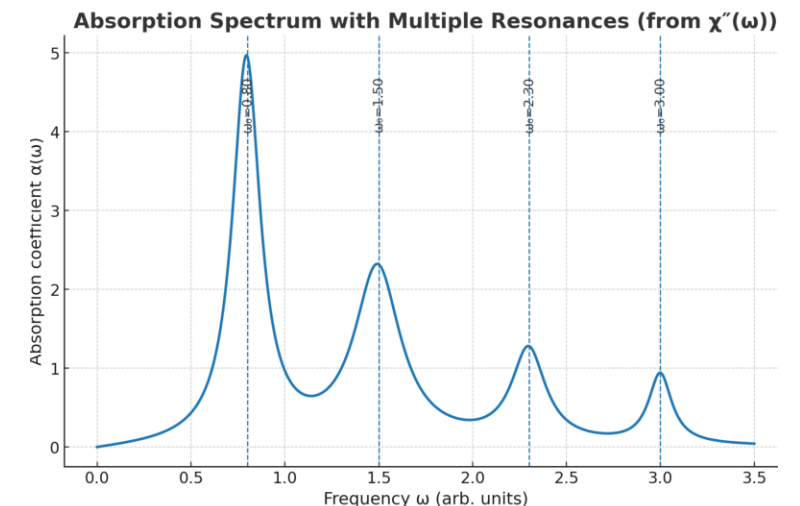
$$\chi^{(1)}(\omega) = \chi'(\omega) + i\chi''(\omega)$$

Real part tells us about the phase velocity change of light

Imaginary part tells us about photon absorption

$$v_p = \frac{c}{n'} = \frac{c}{\sqrt{1 + \chi'(\omega)}}$$

Medium	Refractive Index n	Phase Velocity v_p (m/s)
Air	1.000	3.00×10^8
Water	1.330	2.26×10^8
Glass (BK7)	1.500	2.00×10^8
Diamond	2.400	1.25×10^8



Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

Describes how a material's **polarization** responds **quadratically** to the applied electric field.

$$\mathbf{P}^{(2)}(\omega_3) = \varepsilon_0 \chi^{(2)}(\omega_3; \omega_1, \omega_2) : [\mathbf{E}(\omega_1) \otimes \mathbf{E}(\omega_2)]$$

$$\chi^{(2)}(\omega_3; \omega_1, \omega_2)$$

$\omega_1, \omega_2 \rightarrow$ the input light frequencies

$\omega_3 = \omega_1 \pm \omega_2 \rightarrow$ the output frequency generated.

Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

Describes how a material's **polarization** responds **quadratically** to the applied electric field.

$$\mathbf{P}^{(2)}(\omega_3) = \varepsilon_0 \chi^{(2)}(\omega_3; \omega_1, \omega_2) : [\mathbf{E}(\omega_1) \otimes \mathbf{E}(\omega_2)]$$

In Simple word...

$\chi^{(2)}$ tells us how efficiently, 2 photons (ω_1 and ω_2) can generate an output at ω_3 .

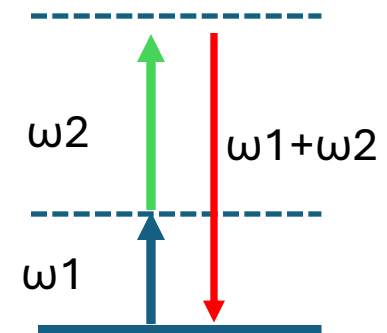
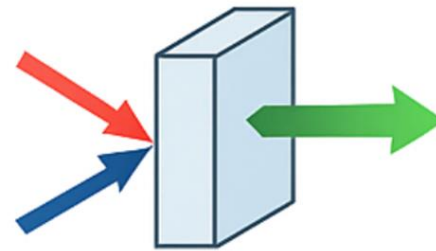
Related to important NLO effects

SHG: $\chi^2(2\omega; \omega, \omega)$

SFG $\chi^2(\omega_1+\omega_2; \omega_1, \omega_2)$

DFG: $\chi^2(\omega_1-\omega_2; \omega_1, -\omega_2)$

EO effect: $\chi^2(\omega; \omega, 0)$



Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

Describes how a material's **polarization** responds **quadratically** to the applied electric field.

$$\mathbf{P}^{(2)}(\omega_3) = \varepsilon_0 \chi^{(2)}(\omega_3; \omega_1, \omega_2) : [\mathbf{E}(\omega_1) \otimes \mathbf{E}(\omega_2)]$$

In Simple word...

$\chi^{(2)}$ tells us how efficiently, 2 photons (ω_1 and ω_2) can generate an output at ω_3 .

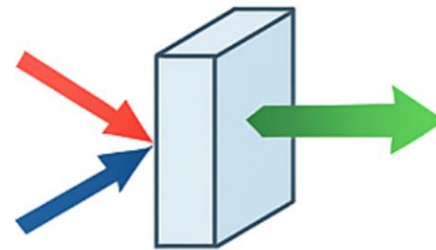
Related to important NLO effects

SHG: $\chi^2(2\omega; \omega, \omega)$

SFG $\chi^2(\omega_1+\omega_2; \omega_1, \omega_2)$

DFG: $\chi^2(\omega_1-\omega_2; \omega_1, -\omega_2)$

EO effect: $\chi^2(\omega; \omega, 0)$



Key idea:
TWO photons
combine or **split** via
the nonlinear
polarization.

Polarization Expansion Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

Advanced Applications of second-order
NLO Effects

Electro-Optic
Modulators (Pockels
Effect): fast quantum
gates and optical
switches.

Spontaneous
Parametric Down-
Conversion (SPDC):
entangled photon pair
sources

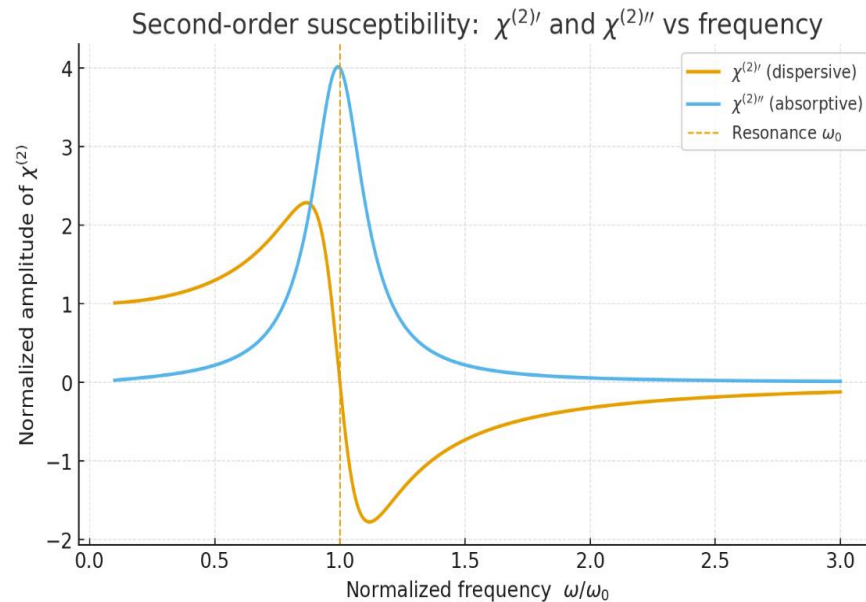
Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

$$\chi^{(2)} = \chi^{(2)'} + i \chi^{(2)''}$$



Conversion efficiency and dispersion

How well light mixes

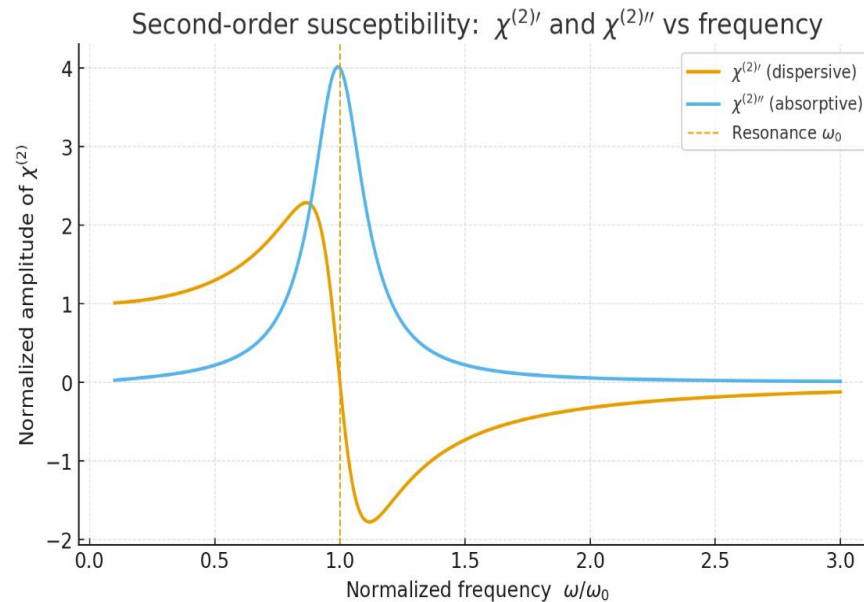
Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

$$\chi^{(2)} = \chi^{(2)'} + i \chi^{(2)''}$$



**Nonlinear Absorption-
Resonance Loses**

**How much light is lost
during the process**

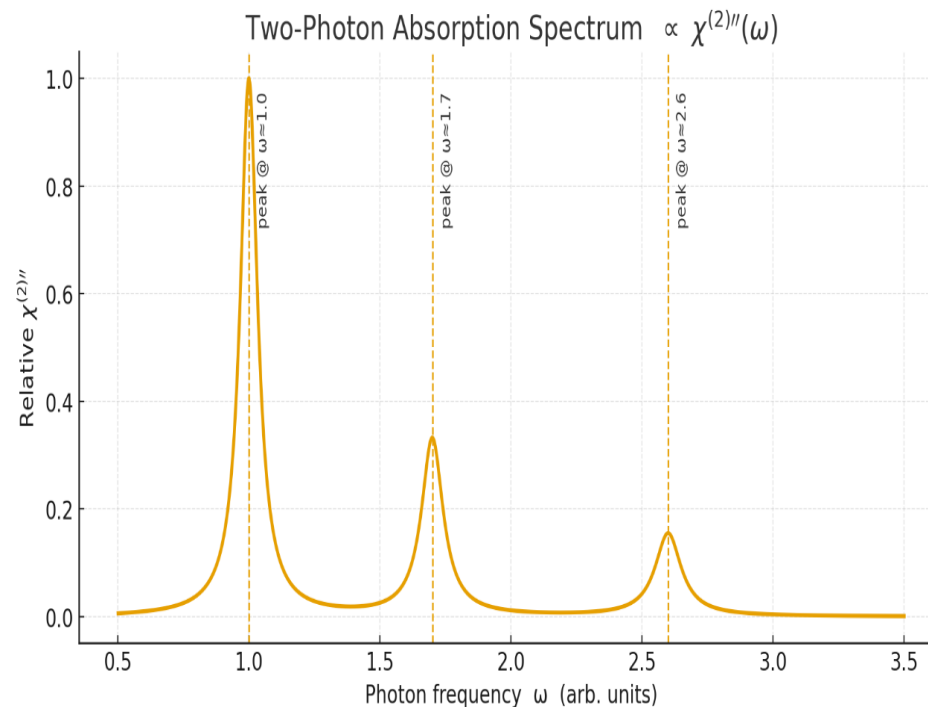
Polarization Expansion

Key points

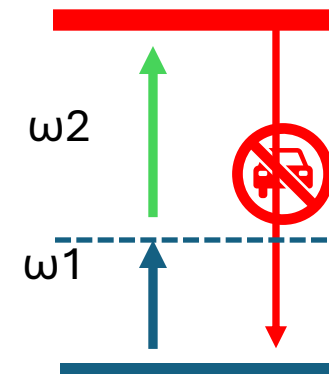
$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(2)}$
Second-order
susceptibility

$$\chi^{(2)} = \chi^{(2)'} + i \chi^{(2)''}$$



In Tells us when and if **two photons** are absorbed **simultaneously**



This happen when $\omega_2 + \omega_1$ matches **a *real electronic state*** of the material

Polarization Expansion

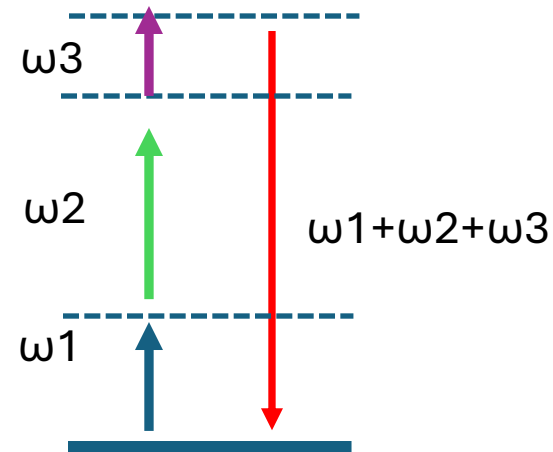
Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(3)}$
Third-order
susceptibility

$$\mathbf{P}^{(3)}(\omega_4) = \varepsilon_0 \chi^{(3)}(\omega_4; \omega_1, \omega_2, \omega_3) \vdots [\mathbf{E}(\omega_1) \otimes \mathbf{E}(\omega_2) \otimes \mathbf{E}(\omega_3)]$$

Describes **nonlinear effects involving three photons**



THG
Creation of new
light

Polarization Expansion

Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(3)}$
Third-order
susceptibility

$$\mathbf{P}^{(3)}(\omega_4) = \varepsilon_0 \chi^{(3)}(\omega_4; \omega_1, \omega_2, \omega_3) \vdots [\mathbf{E}(\omega_1) \otimes \mathbf{E}(\omega_2) \otimes \mathbf{E}(\omega_3)]$$

Similar in principle to $\chi^{(2)}$



- Three-photon absorption (3PA)
- Optical Kerr effect
- Self-phase modulation (SPM)
- Four-wave mixing (FWM)

$\chi^{(3)}$ **universal response**

Always non-zero for

- ✓ Centrosymmetric materials
- ✓ Noncentrosymmetric materials

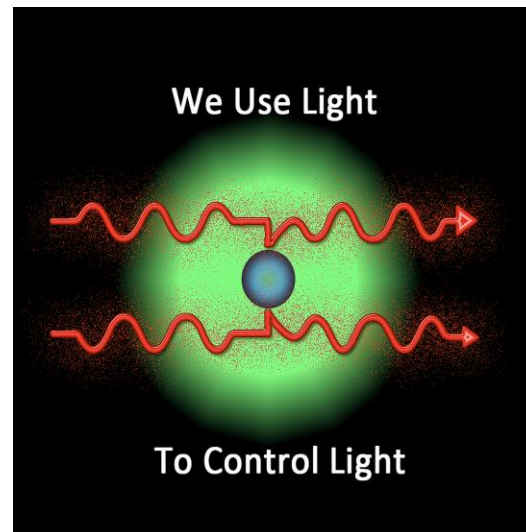
Polarization Expansion Key points

$$P(t) = \varepsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

$\chi^{(3)}$
Third-order
susceptibility

One of the most fascinating NLO
effects

Optical Kerr Effect



Via the refractive Index
of a material



To manipulate
Light

Optical Kerr Effect how it works

To understand how Kerr effect works one has to keep in mind that....

Polarization
Expansion
Key
points

The **refractive index is not fixed**
on the contrary
it **depends on the intensity of the applied optical field.**

$$n(t) = n_0 + n_2 I(t)$$

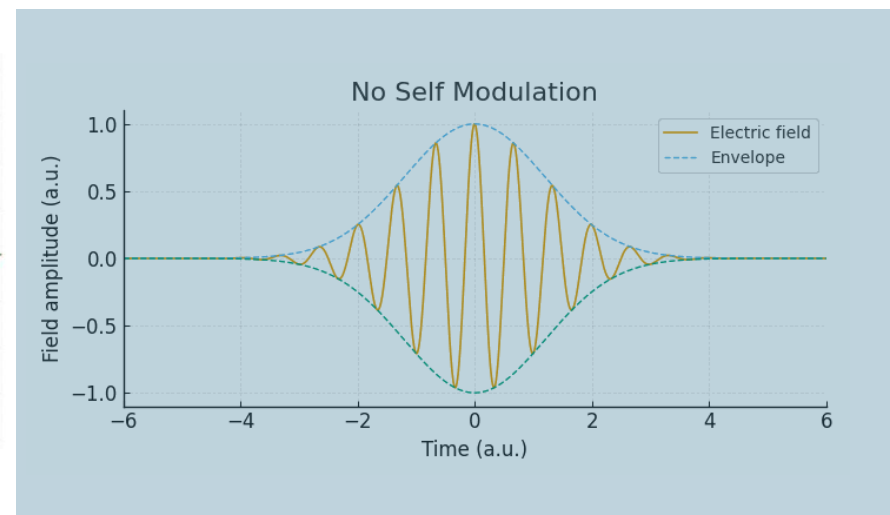
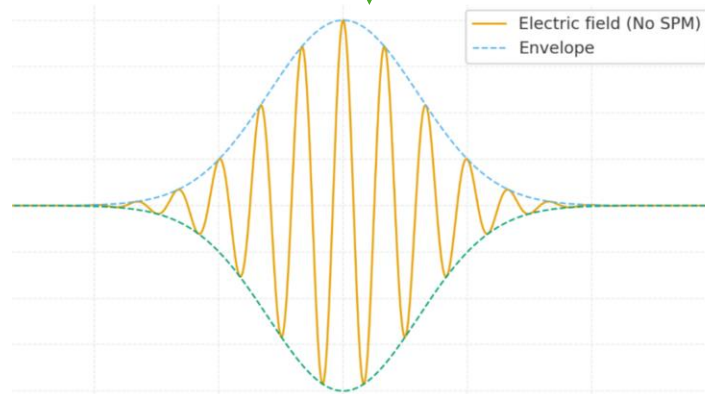
- n_0 : the ordinary (linear) refractive index
- $n_2 I$: the intensity-dependent correction, arising from $\chi^{(3)}$ nonlinear polarization
- For most materials χ^3 is **weak**, so the refractive index seems **static**
- The **Kerr effect** appears only under **intense fields** and when χ^3 is **large**

Optical Kerr Effect how it works

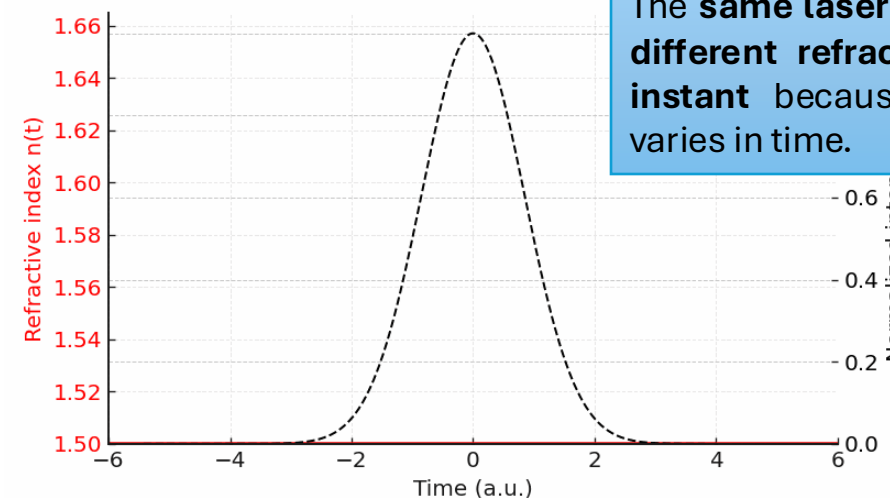
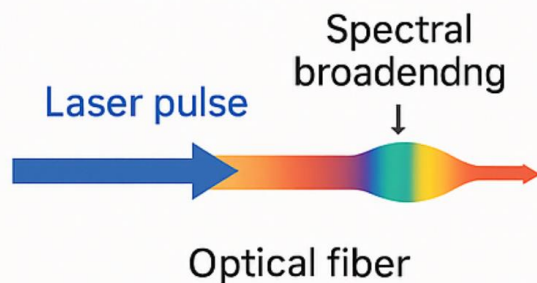
Example a Laser pulse pass through a third-order material

Polarization
Expansion
Key
points

Pulse envelope



Self-Phase Modulation (SPM)

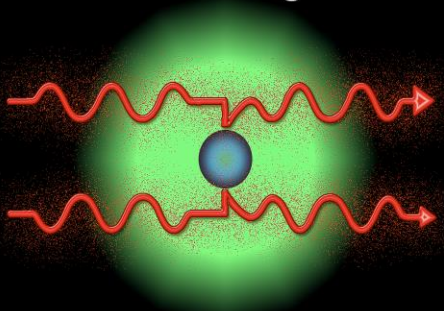


The **same laser pulse** experiences a **different refractive index at each instant** because its own intensity varies in time.

Optical Kerr Effect

One of the most fascinating NLO effects

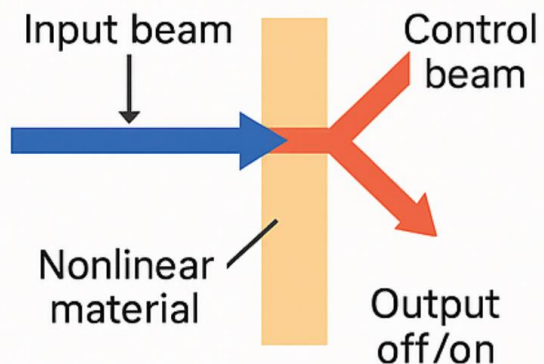
We Use Light



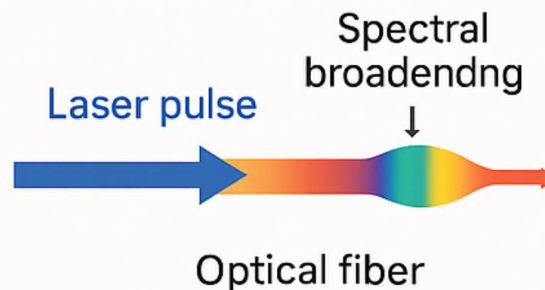
To Control Light

$\chi^{(3)}$
Third-order susceptibility

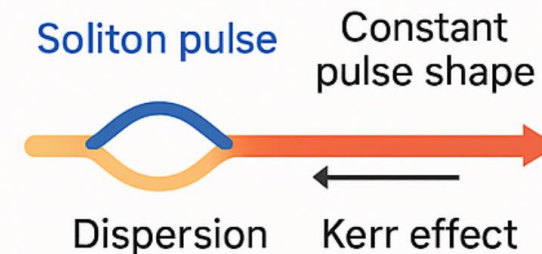
All-Optical Switching



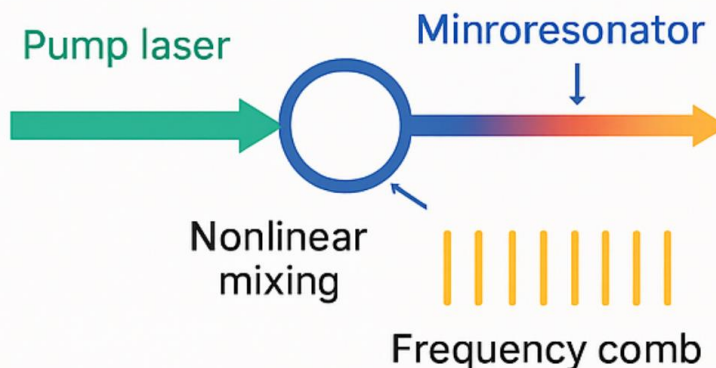
Self-Phase Modulation (SPM)



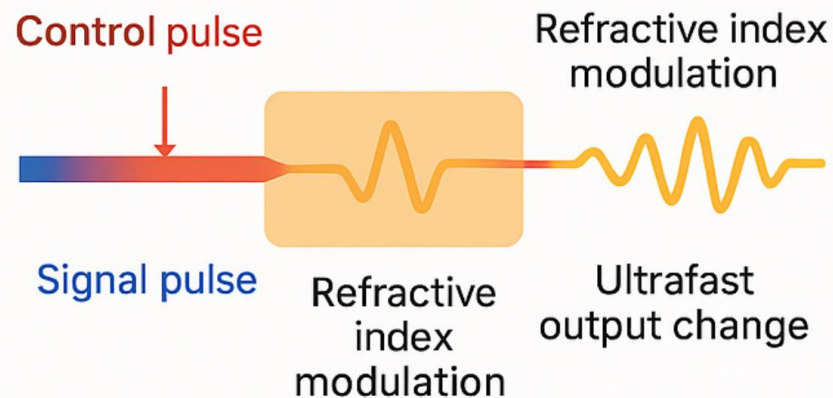
Soliton Propagation



Kerr Frequency Comb in Microresonator



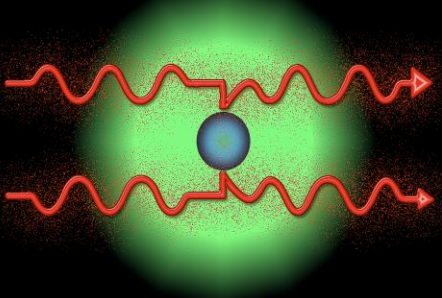
Ultrafast Optical Modulation



Optical Kerr Effect

One of the most fascinating
NLO effects

We Use Light



To Control Light

$\chi^{(3)}$
Third-order
susceptibility

All-Optical Switching

- ✓ Photonic computing (logic gates, memory)
- ✓ Quantum communication (fast routing)
- ✓ All-optical signal processing in telecom networks

Self-Phase Modulation(SPM)

- ✓ Ultrafast pulse shaping and compression
- ✓ Optical coherence tomography (OCT) light sources
- ✓ Supercontinuum generation (white-light sources)

Soliton Propagation

- ✓ Long-distance Optical communication
- ✓ Data Storage
- ✓ Mode-locked lasers (stabilized ultrashort pulses)

Kerr Frequency Comb in Microresonator

- ✓ Ultrafast pulse shaping and compression
- ✓ Optical coherence tomography (OCT) light sources
- ✓ Supercontinuum generation (white-light sources)

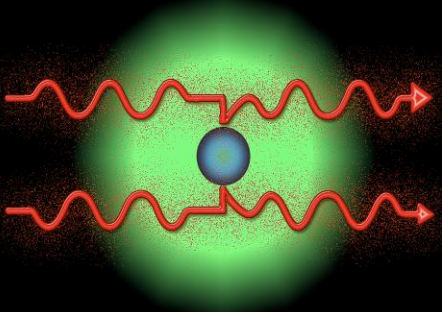
Ultrafast Optical Modulation

- ✓ Real-time optical signal processing
- ✓ On-chip ultrafast modulators for integrated photonics
- ✓ Quantum information processing (ultrafast qubit control)

Optical Kerr Effect

One of the most fascinating
NLO effects

We Use Light



To Control Light

$\chi^{(3)}$
Third-order
susceptibility

All-Optical Switching

- ✓ Photonic computing (logic gates, memory)
- ✓ Quantum communication (fast routing)
- ✓ All-optical signal processing in telecom networks

Self-Phase Modulation(SPM)

- ✓ Ultrafast pulse shaping and compression
- ✓ Optical coherence tomography (OCT) light sources
- ✓ Supercontinuum generation (white-light sources)

Soliton Propagation

- ✓ Long-distance Optical communication
- ✓ Data Storage
- ✓ Mode-locked lasers (stabilized ultrashort pulses)

$$n(t) = n_0 + n_2 I(t)$$

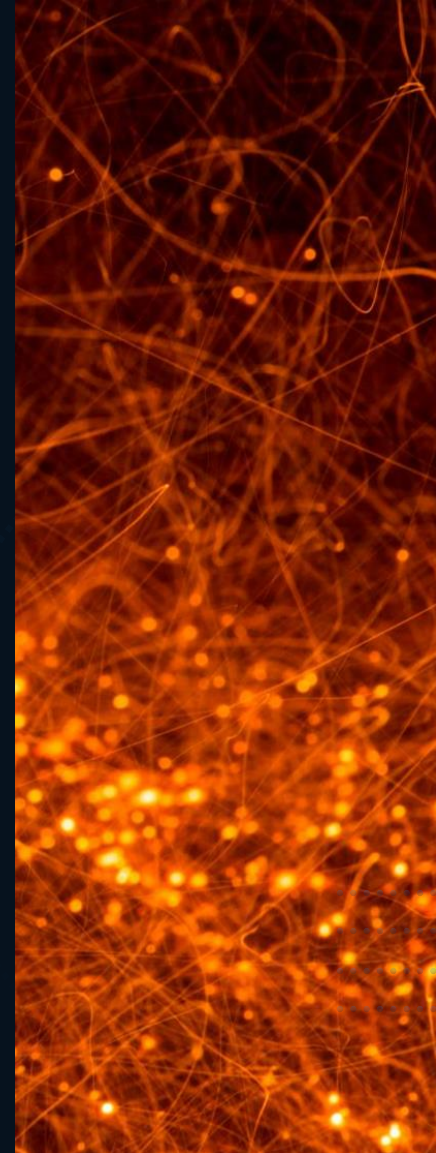
Kerr Frequency Microresonator

- ✓ Ultrafast pulse shaping and compression
- ✓ Optical coherence tomography (OCT) light sources
- ✓ Supercontinuum generation (white-light sources)

Optical Modulation

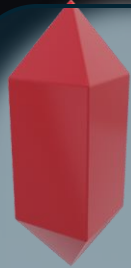
- ✓ Real-time optical signal processing
- ✓ On-chip ultrafast modulators for integrated photonics
- ✓ Quantum information processing (ultrafast qubit control)

Modeling of NLO Phenomena



Modeling of NLO Phenomena

What insights can we gain from modeling
nonlinear optics?



Material X

which frequencies
can be generated

NLO strengths
Conversion efficiency

Spectra behavior
(dispersion-
absorption)

WAVE MIXING

**Refractive Index
change**

**Phase shifts of the
transmitted light**

Response speed

EO Effects (Pockels-Kerr)

Modeling of NLO Phenomena

What insights can we gain from modeling
nonlinear optics?



Material X

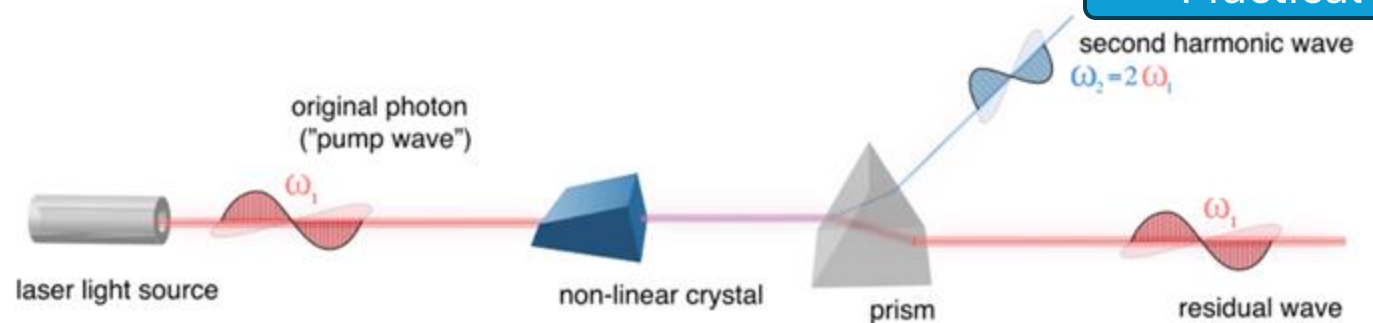
Second Harmonic Generation (SHG):

- Whether the material exhibits SHG behavior
- How strong the SHG signal is
- Which input frequencies stimulate SHG most efficiently

Fundamental

Essential

Practical



Modeling of NLO Phenomena

To get these insights we to construct the general nonlinear Helmholtz equation) :

$$[\nabla^2 + k^2(\omega)] \mathbf{E}(\mathbf{r}, \omega) = -\frac{\omega^2}{\epsilon_0 c^2} \mathbf{P}_{NL}(\mathbf{r}, \omega)$$



Modeling of NLO Phenomena

To get these insights we need construct the general nonlinear Helmholtz equation :

$$[\nabla^2 + k^2(\omega)] \mathbf{E}(\mathbf{r}, \omega) =$$

Left-hand side

- ∇^2 : Laplacian operator $\rightarrow$ describes spatial curvature of the wave.
- $k(\omega)$: wavevector magnitude at frequency ω ,

$$k(\omega) = \frac{n(\omega) \omega}{c},$$

where $n(\omega)$ is the refractive index at that frequency.

- $\mathbf{E}(\mathbf{r}, \omega)$: electric field at frequency ω , as a function of position $\mathbf{r} = (x, y, z)$.

Modeling of NLO Phenomena

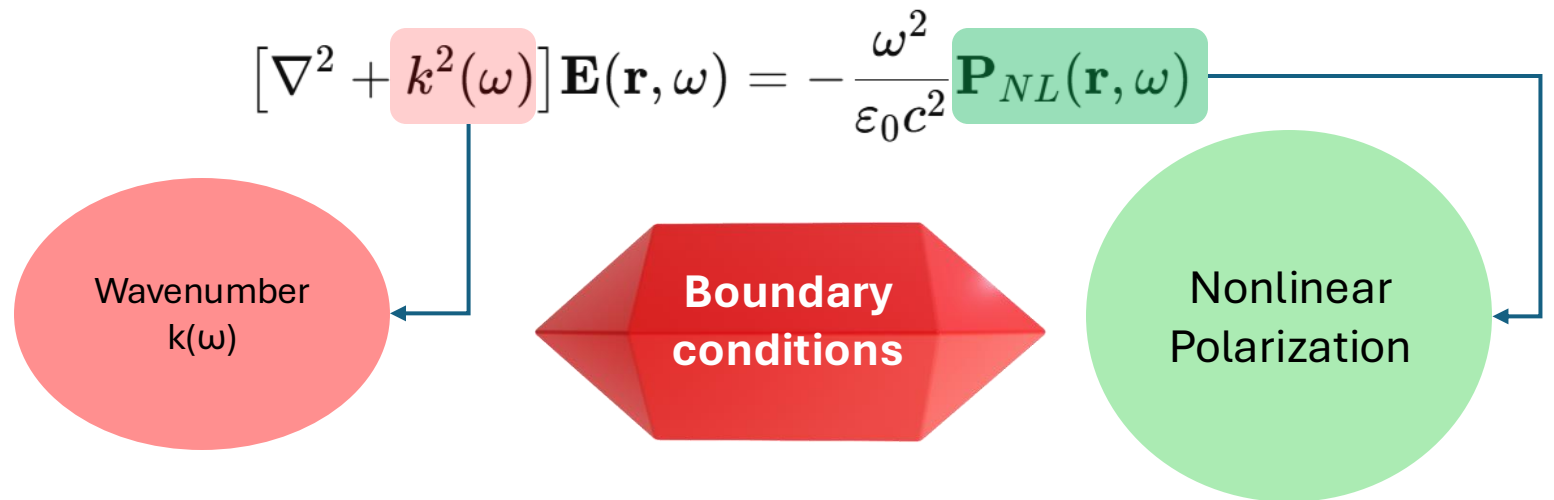
To get these insights we to construct the general nonlinear Helmholtz equation:

$$= -\frac{\omega^2}{\epsilon_0 c^2} \mathbf{P}_{NL}(\mathbf{r}, \omega)$$

- $\mathbf{P}_{NL}(\mathbf{r}, \omega)$: nonlinear polarization at frequency ω , the **source term** that generates new light waves.

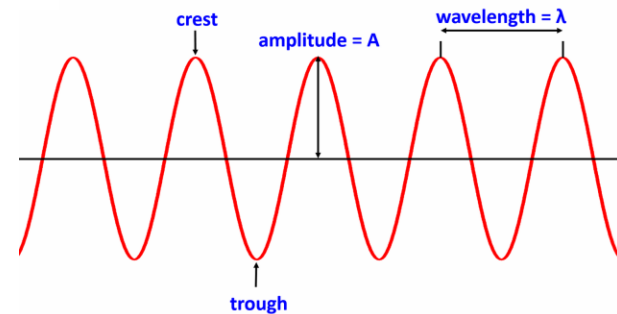
Modeling of NLO Phenomena

To get these insights we to construct the general nonlinear Helmholtz equation:



$$k(\omega) = \frac{\omega}{c} \sqrt{1 + \chi^{(1)}(\omega)}.$$

tells us how closely the crests of a wave are packed inside a material at frequency ω .



Modeling of NLO Phenomena

To get these insights we need to solve the wave equation for nonlinear optics:

$$[\nabla^2 + k^2(\omega)] \mathbf{E}(\mathbf{r}, \omega) = -\frac{\omega^2}{\epsilon_0 c^2} \mathbf{P}_{NL}(\mathbf{r}, \omega)$$

General expression:

$$\mathbf{P}_{NL}(\omega) = \epsilon_0 \sum_{n \geq 2} \chi^{(n)} E^n(\omega).$$

The coefficients $\chi^{(n)}$ are the **nonlinear susceptibilities**.

Classically

$\chi^{(n)} \rightarrow$ material parameters
(numbers or tensors).

Quantum Mechanics

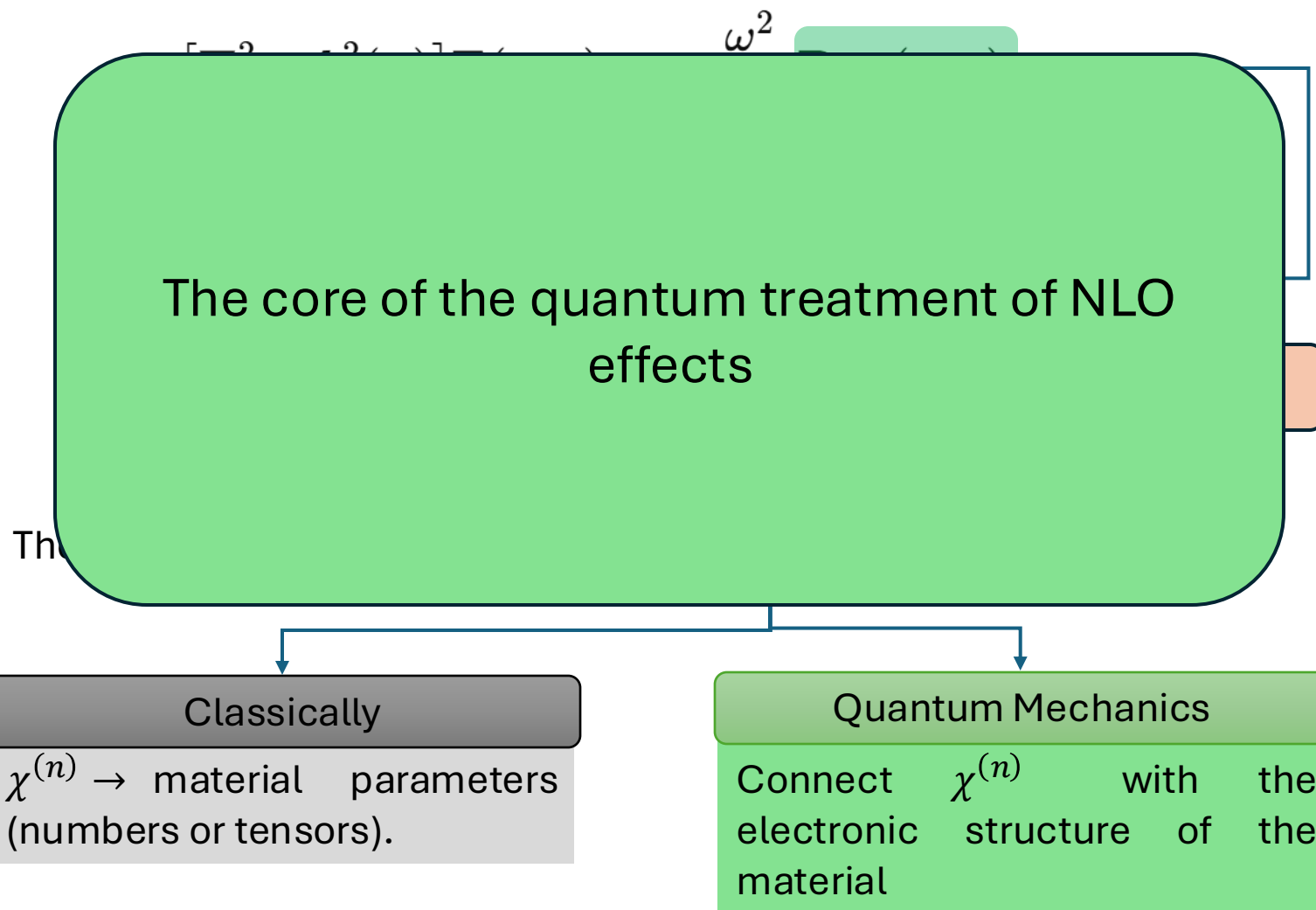
Connect $\chi^{(n)}$ with the
electronic structure of the
material

Nonlinear
Polarization

The most crucial

Modeling of NLO Phenomena

To get these insights we need to solve the wave equation for nonlinear optics:



Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

The connection comes from time-independent perturbation theory, which:

1. Adds a static field to the Hamiltonian:

$$H = H_0 - \mu \cdot F$$

The applied electric field perturbs the electronic Hamiltonian via the dipole operator μ .

2. Expands the ground-state energy in powers of the field:

$$W(F) = W^{(0)} + W^{(1)}F + W^{(2)}F^2 + W^{(3)}F^3 + \dots$$

Each coefficient $W^{(n)}$ encodes how the external field modifies the electronic ground-state energy.

3. Identifies the coefficients $W^{(n)}$ as sums over excited states (microscopic quantum expressions):

- First-order correction (permanent dipole):

$$W^{(1)} = \langle 0 | \mu | 0 \rangle$$

- Second-order correction (linear response):

$$W^{(2)} = \sum_{m \neq 0} \frac{|\langle 0 | \mu | m \rangle|^2}{E_0 - E_m}$$

- Third-order correction (nonlinear pathways):

$$W^{(3)} = \sum_{m, n \neq 0} \frac{\langle 0 | \mu | m \rangle \langle m | \mu | n \rangle \langle n | \mu | 0 \rangle}{(E_0 - E_m)(E_0 - E_n)}$$

Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

The connection comes from time-independent perturbation theory, which:

4. Relates polarization to the energy derivative:

$$P(F) = -\frac{\partial W}{\partial F}$$

This step converts the energy expansion into a polarization expansion.

5. Expands the polarization and compares with the standard definition:

$$P(F) = \varepsilon_0(\chi^{(1)}F + \chi^{(2)}F^2 + \chi^{(3)}F^3 + \dots)$$

By matching terms of equal power in F , the susceptibilities are identified:

$$\chi^{(1)} = -\frac{2W^{(2)}}{\varepsilon_0}, \quad \chi^{(2)} = -\frac{3W^{(3)}}{\varepsilon_0}, \quad \chi^{(3)} = -\frac{4W^{(4)}}{\varepsilon_0}.$$

Thus, the macroscopic susceptibilities $\chi^{(n)}$ emerge directly from the microscopic perturbation expansion of the ground-state energy.

Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

A more generalized and accurate connection comes from **time-dependent perturbation theory**, which:

1. Adds an oscillating field to the Hamiltonian:

$$H(t) = H_0 - \mu \cdot E e^{-i\omega t} + \text{c.c.}$$

2. Expands the polarization in powers of the applied field:

$$P(t) = \varepsilon_0 \left(\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \dots \right)$$

3. Identifies $\chi^{(n)}$ as sums over excited states:

- Numerators $\rightarrow$ products of transition dipole elements $\langle i|\mu|j\rangle$
- Denominators $\rightarrow$ energy gaps shifted by photon energies

$$(E_m - E_0 - \hbar\omega)$$

4. Explains resonance: denominators shrink near optical transitions $\rightarrow$ enhanced response.
5. Connects microscopic to macroscopic: $\chi^{(n)}$ links quantum structure to observable nonlinear polarization.

Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

General SOS Expression for $\chi^{(n)}$ • Full dependence of $\chi^{(n)}$ on electronic structure

$$\chi^{(n)}(-\omega_\sigma; \omega_1, \dots, \omega_n) \propto \sum_{m_1, \dots, m_n} \frac{\langle 0 | \mu | m_1 \rangle \langle m_1 | \mu | m_2 \rangle \dots \langle m_n | \mu | 0 \rangle}{(E_{m_1} - E_0 - \hbar\omega_1)(E_{m_2} - E_0 - \hbar(\omega_1 + \omega_2)) \dots}$$

- Transition dipoles in the numerator
- Resonance denominators shifted by photon frequencies

◆ Philosophy of SOS

- Perturbation theory projects the field's effect onto the system's natural eigenstates
- Matrix elements → measure the **coupling strength** between states
- Energy denominators → measure the **energetic cost** (accessibility of each virtual state)
- SOS = sum over all virtual pathways, weighted by coupling and cost

TAKE HOME MESSAGE



Modeling of NLO Phenomena

1. Defined by the
Electronic
structure of the
materials

$$\chi^{(n)}$$

3. State energy differences
&
Photon energies
Determine resonant effects

2. Built from
virtual electronic
transition
and
State Energy
differences
+
Photon energies

4. Transition moments
Control their Strength

Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

◆ Ingredients of the SOS Method (for Molecules)

1. Electronic states

The molecular eigenstates: the **ground state** $|0\rangle$ and the **excited states** $|m\rangle, |n\rangle, \dots$

2. Transition dipole moments

Defined as

$$\mu_{ij} = \langle i | \mu | j \rangle$$

They quantify how strongly light couples two states.

3. Energy differences

Denominators in SOS terms:

$$\Delta E_{m0} = E_m - E_0 \quad \text{or} \quad E_m - E_0 - \hbar\omega$$

Represent the **energetic cost** of reaching a virtual state.

4. Photon frequencies

External field frequencies $\omega_1, \omega_2, \dots$

determine how the optical field drives virtual transitions.

Modeling of NLO Phenomena

◆ Connection between $\chi^{(n)}$ and Electronic Structure

◆ Ingredients of the SOS Method (for Solids)

1. Electronic states

Bloch eigenstates labeled by **band index** n and **crystal momentum** k :

$$|n, k\rangle.$$

2. Transition matrix elements

Defined in terms of the **velocity operator** v (or equivalently Berry connection):

$$v_{nm}(k) = \langle n, k | \hat{v} | m, k \rangle$$

They describe how an external field couples electronic bands.

3. Energy differences

Band energy separations at a given k :

$$\Delta E_{nm}(k) = E_n(k) - E_m(k) \quad \text{or with shifts} \quad -\hbar\omega$$

They play the role of denominators in the SOS terms.

4. Photon frequencies

External field frequencies $\omega_1, \omega_2, \dots$

enter denominators and control resonance conditions.

Linear Susceptibility $\chi^{(1)}$

$$\chi_{ij}^{(1)}(-\omega; \omega) = \frac{e^2}{\hbar V} \sum_{\mathbf{k}} \sum_{n,m} \frac{f_{n\mathbf{k}} - f_{m\mathbf{k}}}{E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega} p_{nm}^i(\mathbf{k}) p_{mn}^j(\mathbf{k})$$

Second-Order Susceptibility $\chi^{(2)}$

$$\chi_{ijk}^{(2)}(-\omega_\sigma; \omega_1, \omega_2) = \frac{e^3}{\hbar^2 V} \sum_{\mathbf{k}} \sum_{n,m,l} \frac{(f_{n\mathbf{k}} - f_{m\mathbf{k}})(f_{n\mathbf{k}} - f_{l\mathbf{k}}) p_{nl}^i(\mathbf{k}) p_{lm}^j(\mathbf{k}) p_{mn}^k(\mathbf{k})}{[E_{l\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega_\sigma][E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega_1]} + \text{perms}$$

Third-Order Susceptibility $\chi^{(3)}$

$$\chi_{ijkl}^{(3)}(-\omega_\sigma; \omega_1, \omega_2, \omega_3) = \frac{e^4}{\hbar^3 V} \sum_{\mathbf{k}} \sum_{n,m,l,s} \frac{(f_{n\mathbf{k}} - f_{m\mathbf{k}})(f_{n\mathbf{k}} - f_{l\mathbf{k}})(f_{n\mathbf{k}} - f_{s\mathbf{k}}) p_{ns}^i(\mathbf{k}) p_{sl}^j(\mathbf{k}) p_{lm}^k(\mathbf{k}) p_{mn}^l(\mathbf{k})}{[E_{s\mathbf{k}} - E_{n\mathbf{k}} - \hbar(\omega_1 + \omega_2 + \omega_3)][E_{l\mathbf{k}} - E_{n\mathbf{k}} - \hbar(\omega_1 + \omega_2)][E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega_1]} + \text{perms}$$

Definitions

- V : unit cell volume
- $\mathbf{k}$: wavevector in the Brillouin zone
- n, m : band indices (occupied/unoccupied)
- $E_{n\mathbf{k}}$: band energies
- $f_{n\mathbf{k}}$: occupation factors (0 or 1 at T=0)
- $p_{nm}^\alpha(\mathbf{k})$: momentum matrix element
- $\hbar\omega$: energy of the incoming photon
 - $\hbar$: Planck's constant divided by 2π
 - ω : frequency of the applied optical field

Modeling of NLO Phenomena

How to read the indices of χ

$$\chi_{ij}^{(1)}$$

$$\chi_{ijk}^{(2)}$$

$$\chi_{ijkl}^{(3)}$$

- Each index = a spatial direction (x, y, z)
- Number of indices = order of the process + 1
 - $\chi^{(1)}$: 2 indices
 - $\chi^{(2)}$: 3 indices
 - $\chi^{(3)}$: 4 indices

Modeling of NLO Phenomena

How to read the indices of χ

$$\chi_{ij}^{(1)}$$

$$\chi_{ijk}^{(2)}$$

$$\chi_{ijkl}^{(3)}$$

- First index i : direction of induced polarization P_i .
- Remaining indices ($j, k, l \dots$): directions of the applied fields.
 - $\chi_{xx}^{(1)}$:
Polarization along x when the field is applied along x .
 - $\chi_{xy}^{(1)}$:
Polarization along x when the field is applied along y .

Modeling of NLO Phenomena

How to read the indices of χ

$$\chi_{ij}^{(1)}$$

$$\chi_{ijk}^{(2)}$$

$$\chi_{ijkl}^{(3)}$$

- First index i : direction of induced polarization P_i .
- Remaining indices ($j, k, l \dots$): directions of the applied fields.
- $\chi_{xyz}^{(2)}$:
Polarization along x when the fields are applied along y and z .

Tensorial quantities

$$\chi_{ij}^{(1)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ij}^{(1)} \right] \begin{bmatrix} E_j \end{bmatrix}$$

$$\chi_{ijk}^{(2)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ijk}^{(2)} \right] \begin{bmatrix} E_j E_k \end{bmatrix}$$

$$\chi_{ijkl}^{(3)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ijkl}^{(3)} \right] \begin{bmatrix} E_j E_k E_l \end{bmatrix}$$

- $\chi_{ij}^{(1)}$ is a **3×3 tensor (matrix)**.
- 9 components, reduced by symmetry (isotropic → scalar)
- $\chi_{ijk}^{(2)}$ is a **3×3×3 tensor (cube)**.
- 27 components, many vanish by symmetry (centrosymmetric → all zero)
- $\chi_{ijkl}^{(3)}$ is a **3×3×3×3 tensor (hypercube)**.
- 81 components, reduces to ~21 independent in isotropic media

Tensorial quantities

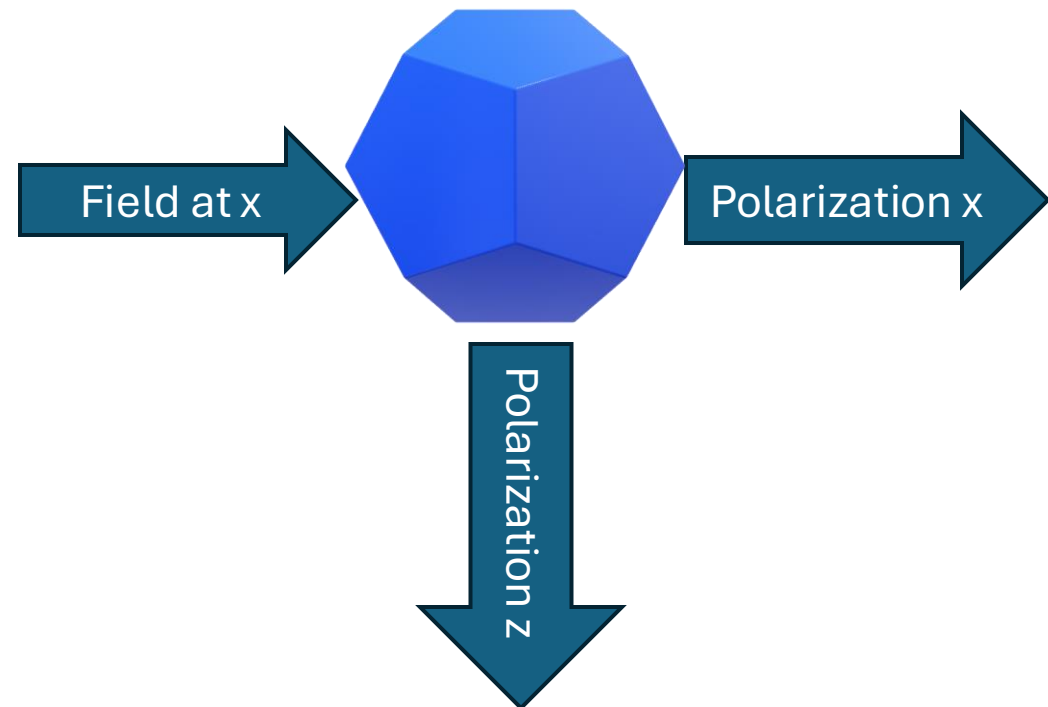
$$\chi_{ij}^{(1)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \epsilon_0 \left[\chi_{ij}^{(1)} \right] \begin{bmatrix} E_j \end{bmatrix}$$

$$\chi_{ijk}^{(2)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \epsilon_0 \left[\chi_{ijk}^{(2)} \right] \begin{bmatrix} E_j E_k \end{bmatrix}$$

$$\chi_{ijkl}^{(3)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \epsilon_0 \left[\chi_{ijkl}^{(3)} \right] \begin{bmatrix} E_j E_k E_l \end{bmatrix}$$

The large number of components simply tell us:

**Any input direction(s)
can contribute to
any output direction.**



Modeling of NLO Phenomena

The real power of SOS is that it reveals *why* nonlinear responses are strong (or weak), giving us a roadmap to design materials with exceptional NLO performance.

- Small band gap $\rightarrow$ small denominators $\rightarrow$ large $\chi^{(n)}$
- Strong dipoles $\rightarrow$ strong coupling $\rightarrow$ large $\chi^{(n)}$
- Near resonance $\rightarrow$ sharp enhancement (but with absorption)
- High density of states $\rightarrow$ more pathways $\rightarrow$ stronger response
- Symmetry rules $\rightarrow$ forbid or allow certain $\chi^{(n)}$

Big nonlinearities = small gaps + strong dipoles + resonant states

Modeling of NLO Phenomena

◆ Limitations of SOS

- To fully exploit SOS, we would need to **solve the time-dependent Schrödinger equation (TDSE)** for the entire molecular or solid system.
- This requires:
 - **All excited states** (energies and wavefunctions)
 - **All transition dipole matrix elements** between them
 - **Frequency-dependent denominators** for every optical process

Modeling of NLO Phenomena

◆ Limitations of SOS

- To fully exploit SOS, we would need to **solve the time-dependent Schrödinger equation (TDSE)** for the entire molecular or solid system.
- This requires:
 - **All excited states** (energies and wavefunctions)
 - **All transition dipole matrix elements** between them
 - **Frequency-dependent denominators** for every optical process
- For realistic molecules and especially **extended solids**, this is **impossible in practice**:
 - The number of excited states is effectively infinite
 - Computing every dipole element is prohibitive
 - Storage and summation over all states is intractable

Modeling of NLO Phenomena

◆ Limitations of SOS

- To fully exploit SOS, we would need to **solve the time-dependent Schrödinger equation (TDSE)** for the entire molecular or solid system.
- This requires:
 - **All excited states** (energies and wavefunctions)
 - **All transition dipole matrix elements** between them
 - **Frequency-dependent denominators** for every optical process

◆ The Practical Route

The most economical way to apply SOS is to use a **single SCF computation**, which provides:

- **Occupied orbitals** → describe the ground-state electrons
- **Virtual orbitals** → approximate excited states
- **Transition dipole integrals** → between occupied and virtual orbitals
- **Orbital energy differences** → used as denominators in SOS formulas

Modeling of NLO Phenomena

◆ Limitations of the Orbital-Based SOS

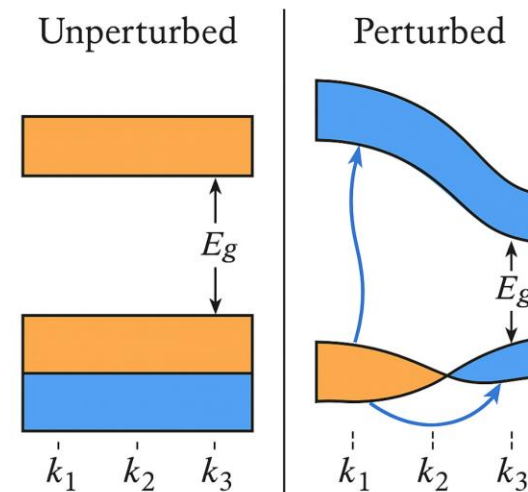
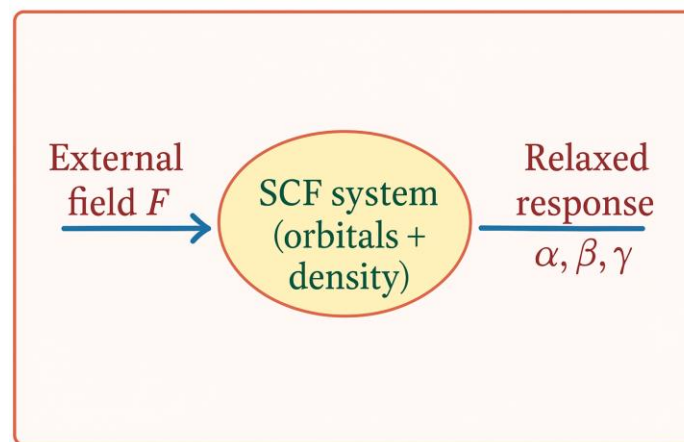
- **Orbital energies $\neq$ true excitation energies**
 - SCF orbital gaps underestimate or overestimate real transition energies.
- **No electron correlation**
 - Excited-state correlation effects are not captured in simple occupied/virtual pictures.
- **Incomplete state description**
 - Virtual orbitals approximate but do not represent real excited states.
- **Double counting or missing states**
 - The orbital basis may miss important many-body excitations or overemphasize others.
- **Resonance positions shifted**
 - Denominators based on orbital energies place resonances at wrong frequencies.

Modeling of NLO Phenomena

CPHF/CPKS

Not a sum over
states

- Compute how orbitals/density relax when a small perturbation (e.g. field) is applied.
- Gives analytic response properties (α , β , γ ...).



Modeling of NLO Phenomena with **CRYSTAL**

NLO Phenomena in CRYSTAL23 – Overview

- $\chi^{(1)}$: Linear Susceptibility
 - Optical constants: refractive index, absorption, dispersion.
 - Systems: molecules, slabs, crystals.
- $\chi^{(2)}$: First Hyperpolarizability (β)
 - Second Harmonic Generation (SHG): frequency doubling.
 - Electro-Optic (Pockels) Effect: field-controlled refractive index.
 - Dispersion of β : frequency-dependent nonlinear response.
 - Systems: molecules, slabs, crystals.
- $\chi^{(2,vib)}$: Vibrational Contributions
 - Enhances SHG and Pockels via ionic displacements.
 - Systems: molecules, slabs, crystals.
- $\chi^{(3)}$: Second Hyperpolarizability (γ)
 - Static Kerr effect (n_2): obtainable from $\chi^{(3)}$.
 - No dynamic $\chi^{(3)}$ effects (e.g., THG) implemented.
 - Systems: molecules, slabs, crystals.

USEFULL TIPS

Modeling of NLO Phenomena with CRYSTAL

- **k-point sampling**
 - Nonlinear susceptibilities converge much more slowly than energies or band structures.
 - $\chi^{(1)}$ and $\chi^{(2)}$ may change significantly when the k-point grid is refined.
 - Always test convergence by progressively increasing k-point density until the NLO tensor elements stop changing appreciably.
- **Basis set choice**
 - Overly diffuse (augmented) primitives can cause near-linear dependence and unstable CPHF/CPKS equations in periodic runs.
 - Recommended practice:
 - Start from CRYSTAL's standard solid-state basis libraries.
 - Add polarization (d, f) functions where needed.
 - Avoid diffuse s/p functions in 2D and 3D cases; they are safe only in finite (0D) molecules.
- **SCF and CPHF/CPKS thresholds**
 - Hyperpolarizabilities are high-order derivatives, very sensitive to residual noise.
 - Use **very tight SCF convergence** and strict integral thresholds.
 - For CPHF/CPKS, reduce residual tolerances to the lowest practical values.

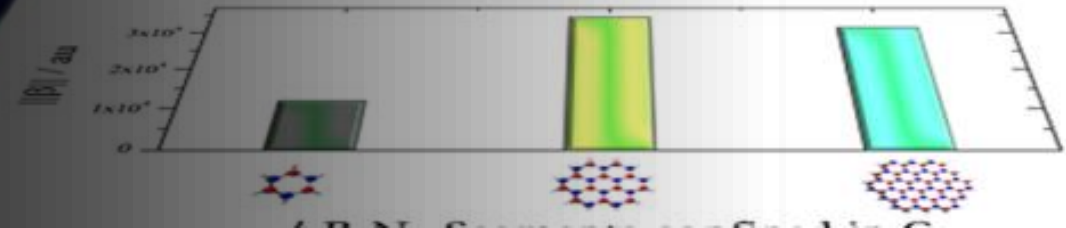
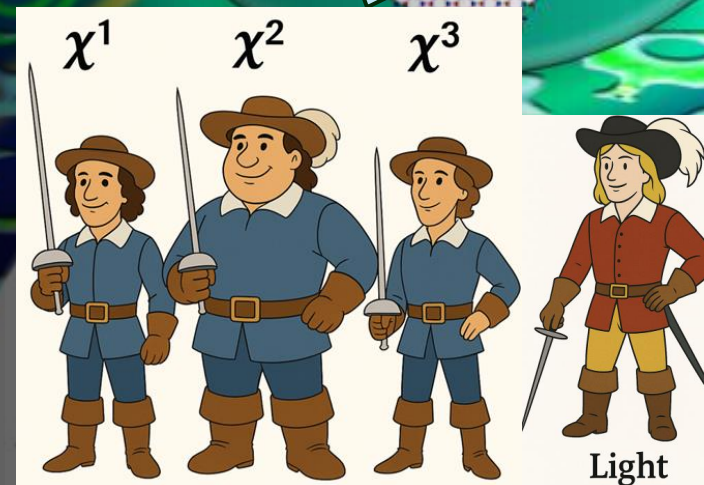


Beyond Linear Responses Exploring NLO Phenomena with CRYSTAL

Panaghiotis Karamanis

Centre National de la Recherche
Scientifique

Thank you for
your
ATTENTION



In the linear response, we explicitly see:

$$\chi_{ij}^{(1)}(-\omega; \omega) \propto \sum_{\mathbf{k}, n, m} \frac{f_{n\mathbf{k}} - f_{m\mathbf{k}}}{E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega} p_{nm}^i(\mathbf{k}) p_{mn}^j(\mathbf{k})$$

- $f_{n\mathbf{k}}$: Fermi–Dirac occupation (0–1 at $T=0$, fractional at finite T).
- The **difference** $f_n - f_m$ ensures the transition only contributes if one state is filled and the other empty.
- In compact χ^2 and χ^3 expressions, occupations are often left **implicit** by assuming the “initial” state n is in the **occupied set**.
- The rigorous form **does include products of Fermi differences**, e.g.:
 - $\chi^2: (f_n - f_m)(f_n - f_l)$
 - $\chi^3: (f_n - f_m)(f_n - f_l)(f_n - f_s)$

These guarantee:

- No contribution from fully empty bands.
- No contribution from completely filled $\rightarrow$ filled transitions.

In the linear response, we explicitly see:

$$\chi_{ij}^{(1)}(-\omega; \omega) \propto \sum_{\mathbf{k}, n, m} \frac{f_{n\mathbf{k}} - f_{m\mathbf{k}}}{E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega} p_{nm}^i(\mathbf{k}) p_{mn}^j(\mathbf{k})$$

- $f_{n\mathbf{k}}$: Fermi–Dirac occupation (0–1 at $T=0$, fractional at finite T).
- The **difference** $f_n - f_m$ ensures the transition only contributes if one state is filled and the other empty.

3. Physical meaning

- At $T=0$ in insulators/semiconductors:
 - $f = 1$ for valence bands, $f = 0$ for conduction bands.
 - Effectively, sums start with n in valence, while m, l, s can be any.
- At finite T or in metals:
 - f can be fractional, so near-Fermi-level transitions contribute continuously.
 - This is why metals and doped semiconductors need **dense k-point sampling**.

Second-Order Susceptibility $\chi^{(2)}$

$$\chi_{ijk}^{(2)}(-\omega_\sigma; \omega_1, \omega_2) = \frac{e^3}{\hbar^2 V} \sum_{\mathbf{k}} \sum_{n,m,l} \frac{(f_{n\mathbf{k}} - f_{m\mathbf{k}})(f_{n\mathbf{k}} - f_{l\mathbf{k}}) p_{nl}^i(\mathbf{k}) p_{lm}^j(\mathbf{k}) p_{mn}^k(\mathbf{k})}{[E_{l\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega_\sigma][E_{m\mathbf{k}} - E_{n\mathbf{k}} - \hbar\omega_1]} + \text{perms}$$

Optical matrix elements

- $p_{nm}^i(\mathbf{k}) = \langle \psi_{n\mathbf{k}} | \hat{p}_i | \psi_{m\mathbf{k}} \rangle$ — **light-matter coupling strength** between bands; determines **oscillator strength** and **selection rules** (symmetry/parity).
- In **molecules**, SOS formulas use **transition dipole moments**:

$$\mu_{nm}^\alpha = \langle \psi_n | \hat{r}_\alpha | \psi_m \rangle.$$

- In **solids**, Bloch states $\psi_{n\mathbf{k}}$ are delocalized $\rightarrow$ position operator $\hat{r}$ is not well defined.
 - Replace with the **momentum (velocity) operator**.

Definition:

$$p_{nm}^\alpha(\mathbf{k}) = \langle \psi_{n\mathbf{k}} | \hat{p}_\alpha | \psi_{m\mathbf{k}} \rangle$$

- Directly determines **oscillator strength** of optical transitions.
- Plays the **same role** as transition dipole moments in molecules.

Tensorial quantities

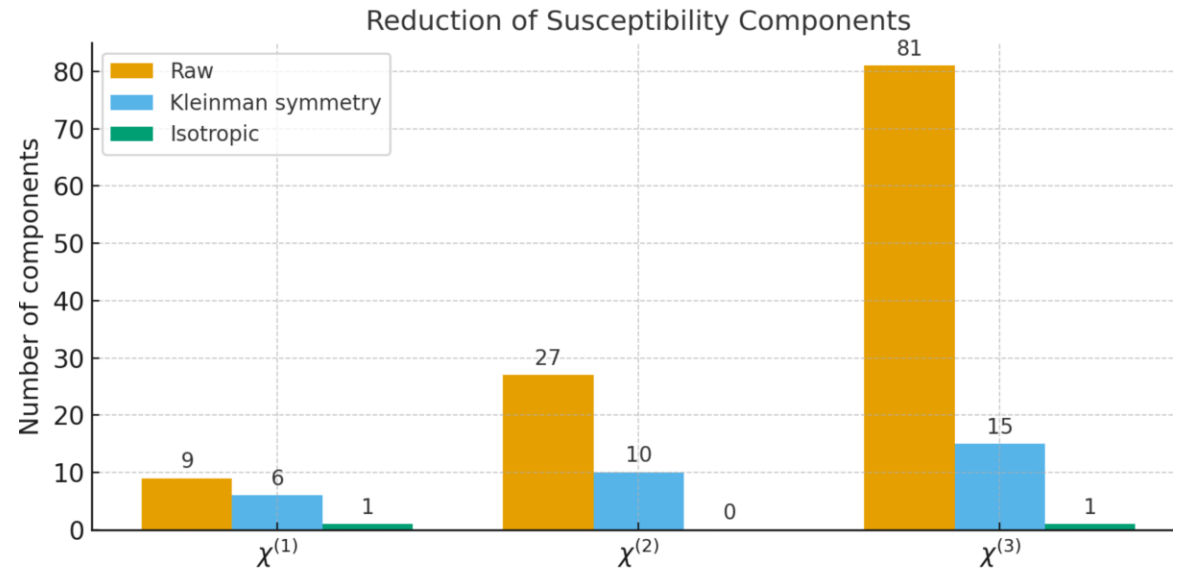
$$\chi_{ij}^{(1)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ij}^{(1)} \right] \begin{bmatrix} E_j \end{bmatrix}$$

$$\chi_{ijk}^{(2)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ijk}^{(2)} \right] \begin{bmatrix} E_j E_k \end{bmatrix}$$

$$\chi_{ijkl}^{(3)} \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \varepsilon_0 \left[\chi_{ijkl}^{(3)} \right] \begin{bmatrix} E_j E_k E_l \end{bmatrix}$$

Kleinman Symmetry (Static / Off-Resonant Case)

- All interacting frequencies **equal** or ≈ 0
- Susceptibility tensor is **invariant under index permutation**
 - e.g.
 $\chi_{xyz}^{(2)} = \chi_{xzy}^{(2)} = \chi_{yxz}^{(2)} = \dots$





$\chi^{(1)}$
Linear susceptibility

$\chi^{(2)}$
Second-order
susceptibility

$\chi^{(3)}$
Third-order
susceptibility

$$\chi^{(2)} = \chi^{(2)'} + i \chi^{(2)''}$$

$$\chi^{(2)}(2\omega; \omega, \omega) = \frac{C}{(\omega_0^2 - \omega^2 - i \gamma \omega)^2},$$

$$\chi^{(2)'}(2\omega; \omega, \omega) = C \frac{(\omega_0^2 - \omega^2)^2 - (\gamma \omega)^2}{[(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2]^2},$$

$$\chi^{(2)''}(2\omega; \omega, \omega) = C \frac{2 (\omega_0^2 - \omega^2) (\gamma \omega)}{[(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2]^2}.$$

$$\chi'(\omega) = \frac{Ne^2}{\epsilon_0 m^*} \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2}$$

$$\chi''(\omega) = \frac{Ne^2}{\epsilon_0 m^*} \frac{\gamma \omega}{(\omega_0^2 - \omega^2)^2 + (\gamma \omega)^2}$$

Modeling of NLO Phenomena

$$\chi^{(n)}$$

Two Ways to Express Susceptibilities from Electronic Features

- Time-independent perturbation theory (static SOS)
 - Treats the system under a static external field.

Time-independent
Schrodinger Equation

$$E \psi(\mathbf{r}) = [\hat{H}_0 - \mu \cdot \mathbf{E}_0] \psi(\mathbf{r})$$

- Time-dependent treatment (TDSE / density matrix)
 - Solves the dynamics under an oscillating field.

Time-dependent Schrodinger
Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = [\hat{H}_0 - \mu \cdot \mathbf{E}(t)] \Psi(\mathbf{r}, t)$$

Modeling of NLO Phenomena

Two Ways to Express Susceptibilities from Electronic Features

- Time-independent perturbation theory (static SOS)
 - Treats the system under a static external field.
 - Valid for off-resonant fields, no damping, ground-state occupancy assumed.
- Time-dependent treatment (TDSE / density matrix)
 - Solves the dynamics under an oscillating field.
 - Captures resonance, absorption, saturation, relaxation.

Modeling of NLO Phenomena

1. Break the problem into a periodic unit cell:

Model the infinite solid with periodic boundary conditions
(Bloch states)

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$$



Electronic structure as energy bands
 $E_n(\mathbf{k})$

Calculate
Matrix elements between bands at each $\mathbf{k}$

Modeling of NLO Phenomena

Matrix elements between Bloch states in periodic systems

For Periodic systems we need to obtain

Matrix element of the velocity operator between Bloch states $|n\mathbf{k}\rangle$ and $|m\mathbf{k}\rangle$

$$r_{nm}^i(\mathbf{k}) = \frac{1}{i\omega_{nm}(\mathbf{k})} \left\langle u_{n\mathbf{k}} \left| \frac{\hat{p}_i}{m_e} \right| u_{m\mathbf{k}} \right\rangle, \quad n \neq m$$

The equivalent quantities in molecules are the transition moments between orbitals

$$\mu_{ab} = \langle \psi_a | \hat{\mathbf{r}} | \psi_b \rangle$$

- Integral over all space
- Orbitals localized and normalizable
- Direct use of the position operator $\hat{\mathbf{r}}$

Same role for
Solids but they
are evaluated at
each k point

Optical -
Transitions

SOS expression

Third-order susceptibility ($\chi(3)$):

$$\chi^{(3)}(-\omega_\sigma; \omega_1, \omega_2, \omega_3) = \frac{N}{\hbar^3 \epsilon_0} \sum_{n,m,l} \frac{\mu_{0n} \mu_{nm} \mu_{ml} \mu_{l0}}{(\omega_{n0} - \omega_\sigma - i\Gamma)(\omega_{m0} - \omega_{12} - i\Gamma)(\omega_{l0} - \omega_1 - i\Gamma)} + \text{permutations}$$

From SOS expressions, good NLO materials should have:

- **Large numerators**
 - Strong transition dipole (or velocity) matrix elements.
 - Allowed transitions with high oscillator strength.
- **Favorable denominators**
 - Energy gaps close to photon frequencies $\rightarrow$ resonance enhancement.
 - But not too close $\rightarrow$ avoid strong absorption and damping.
- **Many contributing states**
 - Multiple excited states with sizable matrix elements $\rightarrow$ constructive contributions.
- **High density of oscillators**
 - More molecules or electronic states per volume $\rightarrow$ larger χ .

Quantum description
Light and Matter
Interaction

We need an expression for
the nonlinear polarization
PNL

Time-Dependent Schrödinger
Equation (TDSE)

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \left(H_0 - \mathbf{E}(t) \cdot \mathbf{P} \right) |\Psi(t)\rangle$$

Describes a
system under
a periodic
perturbation



Time dependent
Optical
perturbation

Represents the
Light - Matter
Interaction

By solving this TDSE equation we
can access

$$P(t) = \langle \Psi(t) | \mathbf{P} | \Psi(t) \rangle$$

As an expectation Value of
the Dipole Operator

Time-Dependent Schrödinger Equation (TDSE)

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \left(H_0 - \mathbf{E}(t) \cdot \mathbf{P} \right) |\Psi(t)\rangle$$

We expand the state as a power series in the applied field:

$$|\Psi(t)\rangle = |\Psi^{(0)}\rangle + E(t) |\Psi^{(1)}(t)\rangle + E^2(t) |\Psi^{(2)}(t)\rangle + E^3(t) |\Psi^{(3)}(t)\rangle + \dots$$

- $|\Psi^{(0)}\rangle$: ground state (no field).
- $E(t)|\Psi^{(1)}\rangle$: first-order change in the state.
- $E^2(t)|\Psi^{(2)}\rangle$: second-order change.
- $E^3(t)|\Psi^{(3)}\rangle$: third-order change.

Allows us to determine

$$P(t) = P^{(0)}(t) + P^{(1)}(t) + P^{(2)}(t) + P^{(3)}(t) + \dots$$

$$P^{(0)}(t) = \langle \Psi^{(0)} | \hat{P} | \Psi^{(0)} \rangle,$$

$$P^{(1)}(t) = E(t) \left[\langle \Psi^{(0)} | \hat{P} | \Psi^{(1)} \rangle + \langle \Psi^{(1)} | \hat{P} | \Psi^{(0)} \rangle \right],$$

$$P^{(2)}(t) = E^2(t) \left[\langle \Psi^{(0)} | \hat{P} | \Psi^{(2)} \rangle + \langle \Psi^{(1)} | \hat{P} | \Psi^{(1)} \rangle + \langle \Psi^{(2)} | \hat{P} | \Psi^{(0)} \rangle \right],$$

$$P^{(3)}(t) = E^3(t) \left[\langle \Psi^{(0)} | \hat{P} | \Psi^{(3)} \rangle + \langle \Psi^{(1)} | \hat{P} | \Psi^{(2)} \rangle + \langle \Psi^{(2)} | \hat{P} | \Psi^{(1)} \rangle + \langle \Psi^{(3)} | \hat{P} | \Psi^{(0)} \rangle \right],$$

Convenient because
we can use these expressions to
model each NLO effect separately

in terms of the field
perturbation

Quantum description
Light and Matter
Interaction

$$P(t) = P^{(0)}(t) + P^{(1)}(t) + P^{(2)}(t) + P^{(3)}(t) + \dots$$

$$P(t) = P_0 + P_L(t) + P_{NL}(t)$$

- P_0 : permanent or static polarization
- $P_L(t)$: linear response, proportional to $E(t)$
- $P_{NL}(t)$: nonlinear response, higher-order terms in $E(t)$

wave equation used in nonlinear optics

$$[\nabla^2 + k^2(\omega)]\mathbf{E}(\mathbf{r}, \omega) = -\frac{\omega^2}{\epsilon_0 c^2} \mathbf{P}_{NL}(\mathbf{r}, \omega)$$

$$\mathbf{P}_{NL}(\mathbf{r}, \omega) = \mathbf{P}^{(2)}(\mathbf{r}, \omega) + \mathbf{P}^{(3)}(\mathbf{r}, \omega) + \dots$$

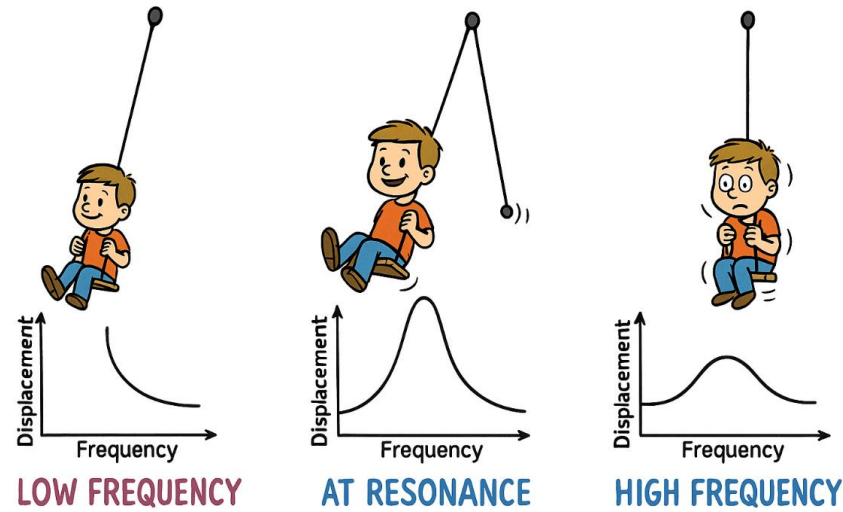
Frequency Domain

Polarization Expansion

Key points

$$P(t) = \epsilon_0 \left[\chi^{(1)} E(t) + \chi^{(2)} E^2(t) + \chi^{(3)} E^3(t) + \dots \right]$$

In principle the polarization expansion **expresses how the electron density of a material responds to an applied optical field (time dependent).**



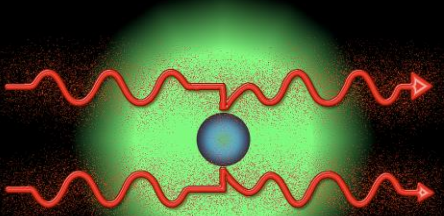
It has its roots in the Lorentz oscillator model.

When the restoring force is harmonic, we get the linear response $\chi(1)$.

When it is anharmonic, higher-order terms naturally appear.

Optical Kerr Effect

We Use Light



To Control Light

One of the most fascinating
NLO effects

$\chi^{(3)}$
Third-order
susceptibility

Intensity of the applied
optical field

Increases /Decreases

The refractive Index of a
material

$$n(I) = n_0 + n_2 I, \quad n_2 \propto \chi^{(3)}$$

n_0 is the linear refractive index,

n_2 is the nonlinear refractive index coefficient

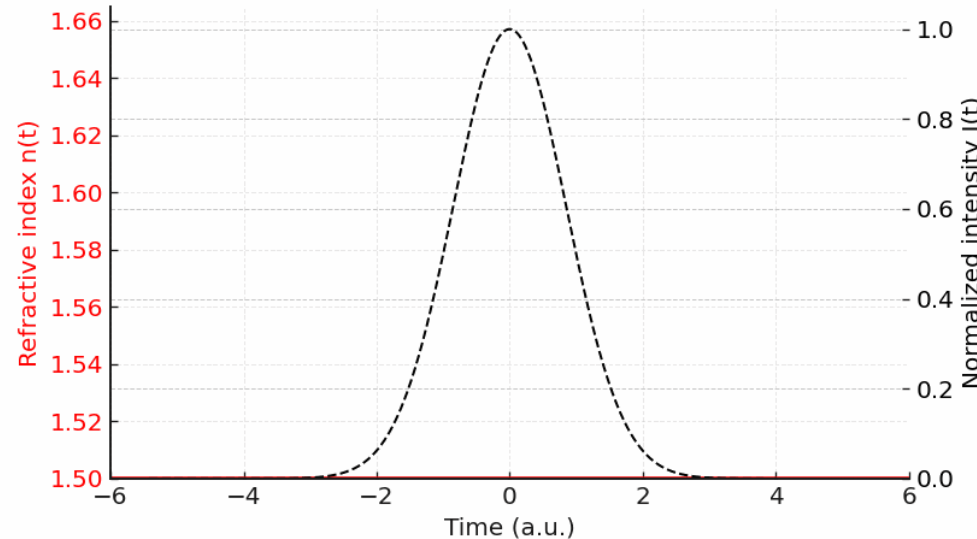
I is the optical intensity.

Time independent

$$n = n_0 + n_2 I$$

Time depended

$$n(t) = n_0 + n_2 I(t)$$





$\chi^{(1)}$
Linear susceptibility

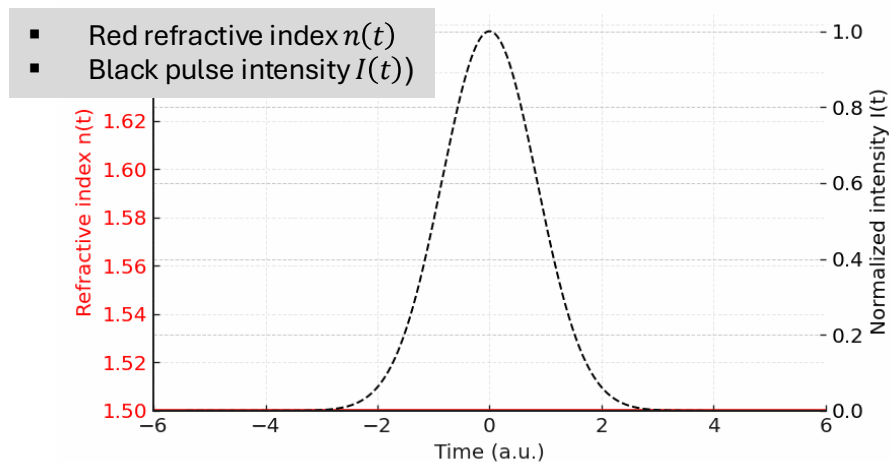
$\chi^{(2)}$
Second-order susceptibility

$\chi^{(3)}$
Third-order susceptibility

Optical Kerr Effect

The refractive index of a medium changes with light intensity

$$n(I) = n_0 + n_2 I, \quad n_2 \propto \chi^{(3)}$$



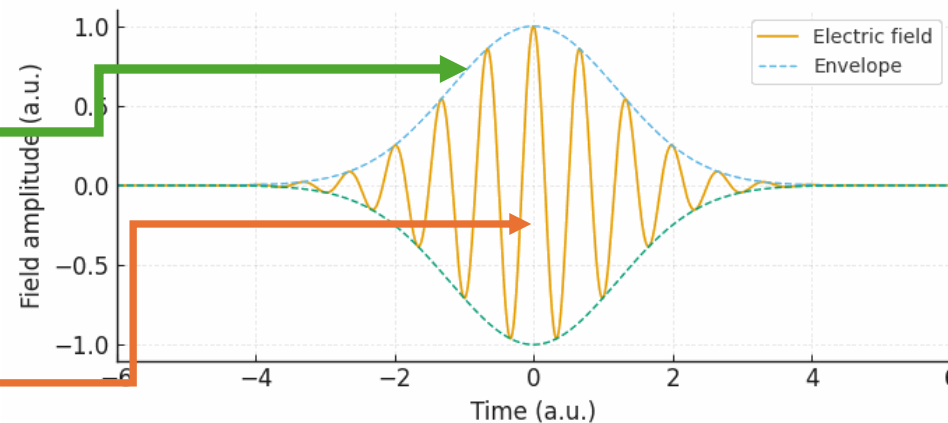
During Kerr effect:

The **same laser pulse** experiences a **different refractive index at each instant** because its own intensity varies in time.



Pulse envelope
Always symmetric

Electric field
oscillations vary with
time.



Laser pulses under the Kerr effect

Light is controlled
with Light
using the Kerr
nonlinearity as the
tool